

Religious Adherence and Local Income: Evidence from U.S. Counties

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Abstract

This paper examines how large, sustained changes in the share of religious adherents affect county-level income in the United States using a stacked difference-in-differences design with inverse propensity score weights. Counties with large declines in adherent shares experience a roughly four percent decline in net income per capita at the time of treatment and a seven percent decline a decade later. The effects of increases in adherents are positive but slower to appear, with income rising by five to seven percent after one to two decades. Part of this relationship reflects changes in county demographics. Increases in adherents are accompanied by increases in the retirement-age population and, in turn, Social Security, Medicare, and asset income. Declines produce the opposite pattern. However, demographics do not explain the full effect. Jobs and net earnings per capita also rise with increases in adherents and fall with declines, and these results hold once we exclude counties with large contemporaneous changes in their retirement-age population.

JEL Codes: Z12, R11, J11, H55

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1 Introduction

What drives differences in economic performance? While scholars have emphasized formal institutions such as property rights and the rule of law (Acemoglu et al., 2001, 2005; Mauro, 1995), others point to culture and social norms as deeper determinants (Doepke and Zilibotti, 2014; Mokyr, 2016; Gorodnichenko and Roland, 2011, 2017). Religion, an enduring and measurable dimension of culture, may shape economic outcomes through its effects on work, education, and social capital. We test these relationships by examining whether large and sustained changes in religious adherents, as a share of the population, across United States counties affect personal income. We then also examine whether these changes are reflected in the aforementioned mechanisms.

The idea that religion could have a direct impact on economic growth was first popularized by the work of Max Weber (1930), and examined empirically by Barro and McCleary (2003).¹ Weber famously argued that the Protestant work ethic (i.e., the belief in hard work as a symbol of Christian faith) enabled the development of capitalism and ultimately led to economic growth. Grier (1997) finds some empirical support for this hypothesis using a sample of 63 former colonies. Barro and McCleary, on the other hand, focus less on the denominational component of religion and more on common religious beliefs and practices (e.g., belief in heaven and hell). Despite the influence of these contributions and the global prevalence of religion, Becker et al. (2024) note that, out of 1,070 pages in the *Handbook of Economic Growth* (Aghion and Durlauf, 2014), religion is mentioned on only four. Thus, much remains to be learned about the influence of religion on economic outcomes.

Within the United States, religion is among the most common civic institutions (Uslaner, 2001) and, as such, may be an important determinant of local development. Yet the existing county-level evidence is limited and mixed. For example, Rupasingha and Chilton (2009) finds that the relationship between religious adherence and county-level

¹While Barro and McCleary (2003) might not have been the first to empirically test the association between religion and growth, the 2003 paper was seminal in that it changed the trajectory of research on the topic.

income varies in sign across denominational groups and specifications, while [Conroy and Deller \(2021\)](#) find that the effects of religion on county-level growth depend partly on the geographic location. Most recently, [Petach and Powell \(2023\)](#) use Barro-style growth regressions in a two-way fixed effects framework and find that counties with higher shares of religious adherents experience *slower* GDP growth. If religious congregations build trust, cooperation, and social capital, as a significant portion of the literature suggests, why is the county-level evidence so weak, and often negative?

We argue that part of the answer lies in what drives changes in the adherent share and in how local economic performance is measured. Religious participation tends to rise with age ([Koenig, 1993](#); [Bengtson et al., 2015](#)), and older households receive more from specific federal programs such as Social Security and Medicare. Moreover, although total income typically falls in retirement, there is evidence that asset income continues to rise through the age of 85 ([Boyle Torrey and Taeuber, 1986](#)). More broadly, retirees tend to draw down their wealth slowly, a pattern referred to as the “retirement savings puzzle” ([French et al., 2023](#)). A county with a growing religious community may therefore also have an aging population receiving more income through assets, Social Security, and Medicare, even as labor force participation falls and output per worker remains unchanged. GDP and other measures of production would capture little of this additional income. Resident income, however, includes income received by county residents regardless of where or how it is generated and would therefore capture much more of it. The economic impact of religion may thus depend on whose income, and which income, one measures.

We examine this idea using large and sustained changes in the share of the population that can be identified as religious adherents across U.S. counties between 1980 and 2010. Rather than regressing the adherent share on growth, like previous work has done, we adopt a quasi-experimental design following the logic of [Grier and Grier \(2021\)](#). Specifically, we identify large and sustained changes in the adherent share, define these changes as a treatment, and trace how net personal income per capita evolves in treated

counties relative to similar never-treated counties.² We separate positive changes (increases in adherents) and negative changes (decreases in adherents) and analyze them separately.³ Our primary estimator is the stacked difference-in-differences estimator of Cengiz et al. (2019), combined with inverse propensity score weighting that balances treated and control counties on pre-treatment income, age structure, education, and labor market conditions.⁴ This design addresses two limitations of the growth-regression framework. First, it does not impose a linear relationship between the level of the adherent share and growth, or rely on a correctly specified functional form for the controls to extrapolate across very different counties. Second, it also traces the dynamic path of outcomes over two decades, including a pre-treatment placebo period when possible.

Relevant theory concerning religion and income offers predictions in both directions. Religious participation may promote trust, cooperation, and pro-social norms that facilitate exchange and community investment (Putnam, 1995; Guiso et al., 2003; Gruber, 2005). Religious participation may also expand social networks, which can facilitate job search and employment through information sharing and referrals (Granovetter, 1973; Montgomery, 1991; Beaman, 2012; Kalfa and Piracha, 2018, e.g.), and church closures have been linked to lower labor force participation and higher unemployment (Kashner and Pienkny, 2026). Religious participation may contribute to human capital accumulation, particularly through its effects on social capital (Coleman, 1988; Freeman, 1986; Gruber, 2005, e.g.), or provide informal social insurance that substitutes for public transfers (Hungerman, 2005; Gruber and Hungerman, 2007). On the other hand, religious participation may divert time and resources from market activity, potentially reducing economic performance even when religious beliefs themselves promote economically productive behaviors (Barro and McCleary, 2003; McCleary and Barro, 2006).

²Net personal income is total personal income less personal current transfer receipts (Higgins et al., 2006; Young et al., 2008). Net personal income does include assets (namely dividends, interest, and rent), which we separate from net earnings in Section 4.

³Details surrounding the construction of our treatments are discussed in Section 2.

⁴We also use entropy-balanced weights as a robustness check. This combination of pre-weighting with a stacked design is now common in the political economy literature (Grier et al., 2025; Bologna Pavlik et al., 2025, e.g.) and can be considered “doubly robust” (Sant’Anna and Zhao, 2020).

The demographic channel, or explanation, we emphasize differs from the more causal factors described above. A change in a county’s age composition can raise resident income through asset income and age-related transfers even if labor force participation and output per worker decline. This is possible because BEA income measures are based on place of residence. They record the income of people who live in a county, including earnings from jobs held elsewhere. BEA’s employment and pay per job, by contrast, are measured by place of work. To better understand what is behind the income response, we first decompose resident income into net earnings, asset income, and individual government transfer programs. We also analyze labor force participation and unemployment rates. All are measured by place of residence. We then turn to place-of-work based measures of employment and pay per job. Finally, we estimate the effects of our treatments on each county’s age structure directly. If the income response is entirely driven by demographic composition, this implies that religion does not have a direct effect on the local economy, even if the local economy is changing along with these demographic changes.

We find that counties experiencing a large, sustained decline in the adherent share see net income per capita fall by roughly four percent at the time of the decline and seven percent a decade later. Counties experiencing a large increase see gains that appear only with a delay: roughly 5.5 percent a decade after treatment for the 2000 cohort and 7.4 percent two decades after treatment for the 1990 cohort. The decomposition reveals two channels, with the first being demographic. After increases, the retirement-age share and median age rise, Social Security, Medicare, and asset income increase immediately, and labor force participation falls. Means-tested transfers are unchanged. Similarly, these same transfer and asset incomes fall after large declines in the share of adherents. The second is an employment channel. Jobs per person increase in response to increases in the adherent share and fall with adherent declines, and net earnings *per capita* follow with a delay, though pay *per job* does not change.⁵ The jobs-per-person and income effects

⁵We also examine educational attainment as a possible mechanism. We find little evidence of changes in educational attainment following changes in adherent shares. These results are reported in Appendix E.

remain when we exclude counties with simultaneous large changes in their retirement-age share during the treatment decade, whereas the decline in transfers after decreases in the adherent share is much smaller. Thus, the employment results do not appear to be driven solely by the demographic changes that accompany shifts in the adherent share.

Our paper makes three contributions. First, we provide quasi-experimental estimates of the effect of large changes in the adherent share on local income, including their dynamics, and find effects that are positive for increases, negative for declines, and robust to alternative treatment thresholds, control groups, and estimators. Second, we show that much of the immediate income response operates through the age composition of the population, through asset income rather than through earnings, accompanied by higher Social Security and Medicare payments. This offers an explanation for the weak and often negative relationships between religion and county output in prior work ([Petach and Powell, 2023](#)). Religious communities can raise resident income without raising productivity. Third, we identify an employment channel that is distinct from this demographic shift. Counties gaining adherents gain jobs relative to their population, and counties losing adherents lose them, with no change in pay per job. Religious communities appear to shape local economies less by making individual workers more productive, as in the tradition following [Weber \(1930\)](#), and more by shaping who lives in a place and how many of them are employed.

The remainder of the paper proceeds as follows. Section 2 describes the data and the construction of the treatment variable. Section 3 details our empirical strategy. Section 4 presents results for income, its components, demographic composition, transfers, and labor market outcomes, and separates the demographic and employment channels. Section 5 concludes.

2 Data

Our sample consists of 2,950 counties within the contiguous United States. We exclude Virginia and the District of Columbia, as well as 24 counties affected by boundary changes or discontinuities in the BEA series during the sample period.⁶ We also exclude two counties that do not appear in every wave of the religion survey. In this section, we first describe the religion data used to construct our treatments, and then define the treatment measures. Second, we describe our outcome variables and the covariates used to construct the weights.

2.1 Religious Adherents

We construct our treatment variable using the share of a county’s population classified as religious adherents, drawn from four waves of the Religious Congregations and Membership Study (or U.S. Religion Census). The Census surveys formal religious bodies and reports the total number of adherents, congregations, membership, and other relevant information for each participating group within the county on a decadal basis. Our focus is on adherents, rather than, e.g., membership, as adherents is a broad concept that is intended to be comparable through time, across space, and across religious groups (U.S. Religion Census, 2026). It includes all members and their children, and other regular participants who are not members.⁷

We use the 1980, 1990, 2000, and 2010 waves, assembled together in a county-level longitudinal file by (Grammich et al., 2018). This file reconciles denominational mergers and schisms, missing data, and changes in membership standards so that information is consistent across waves, better enabling researchers that want to study *changes* in

⁶We exclude Virginia because its independent-city structure resulted in frequent boundary changes over the sample period and the District of Columbia because it is a single consolidated city rather than a county. We identify counties affected by boundary changes using the Census Bureau’s record of substantial changes to counties and county-equivalent entities (see [here](#)) and the footnotes accompanying the BEA’s Table CAINC4.

⁷Focusing just on, for example, membership is problematic because religious group rules differ from one another and change over time, whereas the adherent concept is more uniform (U.S. Religion Census, 2026).

religious characteristics of a county (Bacon et al., 2018).^{8,9} From this data, a county’s adherent share is total adherents, summed across all reporting bodies, divided by the county population the USRC reports alongside those counts.¹⁰

Because the treatment is defined from large changes between consecutive waves, two features of this data pose challenges to identification. First, the set of religious bodies surveyed is not constant, so a county’s measured share can rise simply because a body that did not report in one wave reports (or is created) in the next.¹¹ The data of Grammich et al. (2018), described in Bacon et al. (2018), fill in some of these gaps through interpolation, extrapolation, and by adherent-to-congregation ratios, but those values are estimates. Second, a religious body’s counting practice can change even when it consistently reports throughout. Using adherents rather than membership removes much of this later concern but not all. Fortunately, the longitudinal data provided by Grammich et al. (2018) gives enough information to build an alternative treatment as a robustness exercise. We discuss both our main treatment, and this alternative definition, in turn.

2.2 Treatment and Control: Main Definition

We define a treatment as a large and sustained change in the adherent share between consecutive waves. A “large” change is defined as approximately one standard deviation of the decennial change, or twelve percentage points. A county is treated in a given wave if the ten-year change exceeds 0.12 *and* the following decade’s change does not fall below -0.04 , one third of a standard deviation. A county that loses more than a third of its increase within ten years is therefore not treated. Negative treatments are defined

⁸The Census runs from 1952 to 2020 at roughly ten-year intervals. The 1952 and 1971 waves are not in this file due to lower participation rates and less standardized measurement relative to later collections.

⁹The dataset also corrects for county boundary changes, however we drop these counties from our analysis as there is inconsistency in other data sources.

¹⁰In some years, adherents are double counted as a combined group and their individual component due to schisms and mergers. We use only the combined group in the sum to avoid double counting. For example, the Anglican Church in North America was created in 2009 after seceding from the Episcopal Church. The data includes both groups separately *and* combined in 2010. We use only their combined value.

¹¹Participation expands from 111 bodies in 1980 to 133 in 1990, 149 in 2000, and 236 in 2010, and of the 302 bodies appearing in at least one collection only 48 report in all four decades (Bacon et al., 2018).

symmetrically.¹²

A 12 percentage-point change is a substantial shift in the local presence of organized religion, and importantly, it is plausibly larger than ordinary decennial fluctuations. It is also sustained. This matters because the measured adherent share can move for reasons unrelated to participation. For example, reporting, membership definitions, and denominational coverage could all cause adherent shares to fluctuate. Using this definition, we do not interpret every observed change as meaningful. Rather, we focus on changes large enough to be unlikely products of ordinary measurement variation, and the persistence requirement limits treatment further to changes that remain largely intact a decade later.¹³

Because the definition requires both a preceding *and* a subsequent decadal change, treatment can only be identified in 1990 and 2000. A county cannot be treated in 1980, which would require a comparable 1970 wave, or in 2010, which would require a 2020 wave.¹⁴ Treated counties are assigned to a cohort based on the timing of their *first* qualifying increase. For positive treatments, 147 counties are assigned to the 1990 cohort and 105 to the 2000 cohort, for a total of 252 treated counties.¹⁵ For negative treatments, 53 counties are assigned to the 1990 cohort and 187 to the 2000 cohort, for a total of 240 treated counties.¹⁶ Table 1 summarizes the sample composition by treatment period. Figure 1 plots the mean adherent share of treated counties by decade relative to treatment. Among counties experiencing an increase, the share rises from about 0.57 to 0.74 at treatment and remains above 0.70 two decades later. The corresponding decline is from about 0.78 to 0.56 at treatment, with the share remaining near 0.51 thereafter.

Control counties are those that experience no qualifying increase or decrease at any point, yielding a pool of 2,458 never-treated counties. Thus, the stacked design com-

¹²Analogous methodology and thresholds have been used extensively in previous research. See, e.g., Grier and Grier (2021); Grier et al. (2022); Pavlik and Young (2022); Pavlik and Callais (2025); Bergh et al. (2026).

¹³We report results using slightly higher (0.14) and lower (0.10) thresholds in Appendix A.

¹⁴Although the 2020 Census is available, it is not included in the merged file of Grammich et al. (2018) and therefore cannot be used in our study.

¹⁵Five counties assigned to the 1990 cohort would qualify for treatment again in 2000.

¹⁶Nine counties assigned to the 1990 cohort would qualify for treatment again in 2000.

pares each treatment cohort only with never-treated counties, avoiding the “forbidden comparisons” problem (Callaway, 2023).

2.3 Treatment and Control: Alternative Definition

We additionally use the data notes supplied by Grammich et al. (2018) to construct an alternative treatment indicator as a robustness check (Appendix A). Specifically, we sum adherent information only from religious bodies that appear in all four waves, were never subject in any wave to the missing-data estimation described above, and carry no data note for a change in how adherents are counted. We recalculate each county’s adherent share over these bodies alone. Because the restricted share rests on a smaller set of religious groups, its decennial change is correspondingly less variable, with a standard deviation of (approximately) 0.06, compared to 0.12 for the baseline measure. Thus, we use the smaller threshold here (0.06), and require that the change be sustained over the following decade within one third of this value. This alternative treatment indicator then retains only counties on which it agrees with the original treatment measures. In other words, a county is treated if they experience a qualifying change in both measures in the same wave. This holds for 38 counties for positive treatments and 116 for negative treatments.¹⁷ Similarly, counties are included as controls if neither measure identifies a qualifying increase or decrease at any point.

2.4 Outcome Data

Our primary dependent variable is log *net* personal income per capita at the county level. We construct this measure from the Bureau of Economic Analysis (BEA) Regional Economic Accounts, Table CAINC4 (1969–2022). From this, we extract county level data on total personal income, population, and personal current transfer receipts. Following Higgins et al. (2006); Young et al. (2008), we define net personal income as total personal

¹⁷The negative change captures more movement because there is a growth in the number of groups despite this number being held constant here. Thus, there is some switching across religious groups that these measures ignore. This is why we focus on the overlap only.

income less personal current transfer receipts. We deflate this measure to constant 2017 dollars using the BEA implicit price deflator for gross domestic product (NIPA Table 1.1.9, index 2017 = 100), divide by BEA county population, and take the natural log. By excluding transfer receipts, net personal income more closely captures income generated through productive activity than does total personal income.

Net personal income nonetheless combines two components that the BEA reports separately: net earnings, and dividends, interest, and rent (which we refer to as assets). Because the latter is received largely by older and wealthier households, the two components can respond very differently to a change in a county’s population. Both components, like net income itself, are measured by place of residence. We therefore examine each separately, along with two measures of local labor demand from the same BEA table that are measured by place of work: wage and salary disbursements per (wage and salary) job, and total employment per person. All dollar-valued series are deflated, expressed per capita (or per job), and logged in the same way as net income.

We also examine the personal transfer receipts that we net out of personal income. Using BEA Table CAINC35, we separate government transfers into their major programs: Social Security, Medicare, public assistance medical care (primarily Medicaid), income maintenance programs (Supplemental Security Income, SNAP, the EITC, and related programs), unemployment insurance, and veterans’ benefits. This allows us to separate transfers driven largely by the population’s age structure, especially Social Security and Medicare, from means-tested programs that may respond more directly to local economic conditions or to informal insurance provided by churches.

Our remaining outcomes include the unemployment rate, labor force participation rate, and two measures of the age structure (the population share aged 65 and older and median age). These measures are drawn from the 1980, 1990, and 2000 decennial Census long-form samples and, because the long-form was discontinued after 2000, from the 2006-2010 American Community Survey five-year estimates, accessed through IPUMS NHGIS (National Historical Geographic Information System) ([Manson et al., 2024](#)). Un-

employment and labor force participation allow us to examine whether changes in income operate through labor market outcomes and the age measures allow us to trace changes in population composition directly.

2.5 Covariate Data

Because treated and never-treated counties differ at baseline, we reweight the controls based on a set of county characteristics standard in studies of county income and growth (e.g., [Higgins et al., 2006](#); [Pavlik and Tackett, 2022](#)). All covariates are measured ten years prior to treatment: 1980 characteristics for the 1990 treatment cohort and 1990 characteristics for the 2000 treatment cohort.

The common covariate set, used for every outcome, contains two lags of log net income per capita (ten and twenty years before treatment), the lagged adherent share, age composition (the shares of the population that are children, teenagers, and of working age), the male population share, the urban population share, educational attainment (the shares of adults aged 25 and older with a high school diploma only, and the share with a college degree), the military share of the labor force, the unemployment rate, and the labor force participation rate. These covariates are summarized in [Table 2](#), and [Table 3](#) compares treated and control counties on them before weighting. The age shares are particularly important for our analysis because they balance the pre-treatment demographics across treatment and controls. The demographic changes we observe after treatment, then, do not reflect initial differences in age composition.

In addition to the common covariate set, each specification includes pre-treatment values of its own outcome. For outcomes drawn from the BEA (net income, net earnings, assets, pay per job, jobs per person, and each transfer series) we include values measured ten and twenty years before treatment. For net income, these two lags are already included in the common set of covariates. For Census outcomes, namely the retirement-age share, median age, unemployment, and labor force participation we include the outcome measured ten years before treatment. A twenty-year lag is not available for these out-

comes because our data do not include the 1970 Census, which would be required for the 1990 treatment cohort. Thus, for example, the Social Security specification matches on two lags of Social Security per capita, two lags of net income per capita, and the remaining common covariates. With the exception of income, transfers and their components, and religious adherents, all covariates are drawn from the decennial Census long-form samples, accessed through IPUMS NHGIS (Schroeder et al., 2026).

3 Empirical Strategy

3.1 Stacked Difference-in-Difference Set Up

We estimate the effect of large, sustained changes in the adherent share using a stacked difference-in-differences design following Cengiz et al. (2019). As described above, counties are treated in either 1990 or 2000 depending on when they *first* experience a qualifying change in the adherent share. The 1990 cohort contains 147 positive and 53 negative treatments, while the 2000 cohort contains 105 positive and 187 negative treatments.¹⁸ Positive and negative treatments are estimated separately. Within each cohort, treated counties are compared with counties that never experience a qualifying increase *or* decrease in the adherent share over the sample period. We then stack the cohort-specific samples.

For county i , treatment cohort g , and year t , we estimate

$$Y_{igt} = \alpha_{ig} + \lambda_{gt} + \sum_{k \in \mathcal{K}} \beta_k [1\{i \in \mathcal{T}_g\} \times 1\{t - g = k\}] + \varepsilon_{igt},$$

where Y_{igt} is the outcome of interest, log net personal income per capita in our main specification, α_{ig} denotes county-by-cohort fixed effects, and λ_{gt} denotes cohort-by-year fixed effects. The indicator $1\{i \in \mathcal{T}_g\}$ identifies counties treated in cohort g , while $k = t - g$

¹⁸Because our data on religious adherents begin in 1980, we treat 1980 as a baseline and assume that all counties are untreated at that point. We also present results separately for the 2000 cohort, for which this assumption is unnecessary.

denotes event time. We omit the decade immediately preceding treatment, $k = -10$, as the reference period. Thus, β_0 , β_{10} , and, where available, β_{20} measure the difference in outcome changes between treated and control counties at treatment and in the decades that follow. We estimate the same specification separately for each outcome: net income, its components (net earnings and dividends, interest, and rent), jobs per person and pay per job, the retirement-age share and median age, each transfer series, unemployment, labor force participation, and educational attainment. Standard errors are clustered at the county level throughout.

One complication is that the full event window is not available for both cohorts. For the 1990 cohort, we observe outcomes at $k = \{0, 10, 20\}$, but cannot construct a comparable $k = -20$ observation because doing so would require identifying treatment status using data on adherents prior to 1980. For the 2000 cohort, we instead observe $k = \{-20, 0, 10\}$, with $k = -20$ providing a pre-treatment placebo estimate, but cannot observe $k = 20$ because the 2020 religion data are not included in our longitudinal dataset. Pooling the full event windows would therefore cause cohort composition to change across event time: the $k = -20$ coefficient would be identified only from the 2000 cohort, while the $k = 20$ coefficient would be identified only from the 1990 cohort. This can cause a composition bias in stacked estimates as explained in [Wing et al. \(2024\)](#).

We address this in two ways. First, our pooled specification uses only $k = \{0, 10\}$, with $k = -10$ as the reference period, which are observed for both cohorts. We restrict the estimation sample itself to the common event window, for both treated and control counties, so that the same cohort composition underlies each reported coefficient. The cost of this approach is that there is no additional pre-treatment period with which to assess pre-trends. Second, we estimate each cohort separately over its full available event window: $k = \{0, 10, 20\}$ for the 1990 cohort and $k = \{-20, 0, 10\}$ for the 2000 cohort (both with $k = -10$ as the reference period). The latter allows us to examine whether treated and control counties were already on different trajectories prior to treatment.

3.2 Inverse Propensity Score Weighting

Of course, treatment is not random, and treated counties may differ systematically from never-treated counties. We therefore follow [Grier et al. \(2025\)](#) and pre-weight the control group separately for each treatment-cohort stack.¹⁹ For example, for the 1990 positive-treatment cohort, we construct weights that make the never-treated controls more closely resemble the treated counties on pre-treatment characteristics. We repeat this procedure separately for the 2000 cohort and for each treatment definition.²⁰ Combined with the stacked difference-in-differences design, this approach can be thought of as “doubly robust” ([Sant’Anna and Zhao, 2020](#)). We implement the pre-weighting using both inverse propensity score weighting (IPW) and entropy balancing (EB), although we report and discuss only the IPW estimates in the main text.

Weights are constructed using pre-treatment county characteristics measured ten years prior to the treatment wave. For the 1990 cohort, covariates are measured in 1980; for the 2000 cohort, they are measured in 1990. As described in the previous section, the covariate set is the common set of pre-treatment characteristics, including two lags of net income per capita and the age structure, plus the outcome’s own history where it is not already included. Because the covariate set includes these outcome-specific lags, the propensity model is re-estimated for each outcome. The exception is net income, unemployment, labor force participation, and educational attainment, whose lags are already in the base covariate set, so these five outcomes share the same weights. Table C3 in Appendix C reports standardized differences in the matching covariates before and after weighting. For positive treatments, IPW reduces all differences below 0.02 standard deviations. For negative treatments, IPW leaves differences of up to 0.16 standard deviations, which entropy balancing removes. Entropy-balanced estimates are similar and are reported in Appendix B.

Propensity scores are estimated separately for each treatment cohort and treatment

¹⁹[Bologna Pavlik et al. \(2025\)](#) apply a similar approach in their study of “revolutionary” constitutions.

²⁰Thus, the same never-treated county can receive different weights across cohort stacks if the treated cohorts differ in their pre-treatment characteristics.

type using a logistic regression on this covariate set. We then construct normalized inverse propensity score weights targeting the average treatment effect on the treated (ATT). Following [Wing et al. \(2024\)](#), weights are rescaled within each cohort-specific stack so that the aggregate weight assigned to never-treated controls equals the aggregate weight assigned to treated counties. Each stack therefore contributes to the pooled estimate in proportion to its number of treated counties.

We consider two sets of robustness checks. The first set varies the treatment and control groups. We redefine treatment using thresholds of 0.10 and 0.14, in place of 0.12, with the persistence requirement scaled to one third of each threshold. With the original treatment threshold, we exclude “near-miss” controls, or never-treated counties whose adherent share changed by more than 0.12 in 1990 or 2000 but did not meet the persistence requirement, and test the robustness of our results. Lastly, we use the alternative treatment measure described above, which, because it yields too few treated counties in the 2000 cohort, we estimate for the 1990 cohort only. These results are reported in [Appendix A](#).

The second set of robustness checks varies the estimator. We replace the inverse propensity score weights with entropy-balanced weights while maintaining the stacked difference-in-differences design. We then estimate the stacked specification with only the corrective aggregation weights of [Wing et al. \(2024\)](#), without reweighting the control group on pre-treatment covariates. Lastly, we estimate a conventional two-way fixed effects event study on the unstacked panel. These results are reported in [Appendix B](#).

4 Results

We present the results for each outcome in the same format. In each figure, the left column reports large increases in the adherent share and the right column reports large decreases. The top row reports pooled estimates over the common post-treatment window ($k = \{0, 10\}$). The bottom row shows effects for each cohort separately over their full

event windows: $k = \{0, 10, 20\}$ for the 1990 cohort and $k = \{-20, 0, 10\}$ for the 2000 cohort, with $k = -10$ serving as the reference period. All coefficients underlying the figures are reported in Appendix D.

4.1 Net Income

Figure 2 reports the effects on net income per capita. Declines in the adherent share have an immediate and persistent effect. Net income per capita falls by 4.1 percent at the time of the decline and by 7.0 percent a decade later, and both estimates are statistically significant. The cohort estimates tell the same story: income in the 1990 cohort is 11.1 percent lower at treatment and 9.2 percent lower two decades later (significant at the ten percent level), and income in the 2000 cohort is 7.8 percent lower a decade after treatment. The pre-treatment estimate for the 2000 cohort is small and insignificant (-0.8 percent).

Increases operate with a delay. The pooled estimates are positive but imprecise, 0.9 percent at treatment and 2.3 percent a decade later. The cohort estimates are a bit more informative. Neither cohort's income changes at the time of treatment, but income is 5.5 percent higher a decade later in the 2000 cohort and 7.4 percent higher two decades later in the 1990 cohort, both significant at the one percent level. The 2000 cohort shows no pre-trend.

4.2 Income Components & Jobs

Figure 3 separates net income into its two main components: net earnings per capita and dividends, interest, and rent (assets). Assets respond immediately to increases in the adherent share, rising 3.6 percent at treatment and 4.7 percent a decade later in the pooled estimates. Earnings, by contrast, do not respond at treatment but rise only later, by 6.8 percent a decade after treatment in the 2000 cohort and 6.0 percent two decades after treatment in the 1990 cohort (significant at the ten percent level). The delayed part of the net income response to increases therefore comes from net earnings, while

assets rise immediately. Declines reduce both components significantly throughout the full post-treatment period when looking at the pooled data.

Figure 4 examines the earnings response further using two place-of-work measures. Pay per wage and salary job does not change in either direction as the pooled estimates are small and insignificant at every horizon. However, jobs per person do change, rising by 0.014 a decade after an increase (0.027 in the 2000 cohort), and falling by 0.018 a decade after a decline. The earnings response therefore appears to come from more jobs in the county relative to its population rather than from higher pay per job, although net earnings, which are measured by place of residence, can also reflect earnings from jobs held outside the county.

4.3 Transfers & Demographic Composition

Government transfers are excluded from net income and thus cannot mechanically explain the effects documented above. They are nonetheless informative for two reasons. First, congregations may provide informal support that substitutes for means-tested public assistance, so religion could reduce reliance on those programs. Second, because eligibility for Social Security and Medicare is tied to age, these transfers can reveal changes in a county's age composition. Total government transfers per capita move with the adherent share in both directions, rising by 2.9 percent at treatment and 3.8 percent a decade later after increases and falling by 3.5 and 4.5 percent after declines (Figure 5).

Breaking transfers down by program shows where this response comes from. Social Security and Medicare (per capita) move with the adherent share in both directions (Figure 6). However, means-tested transfers look different (Figure 7). Medicaid does not respond in either direction. Income maintenance programs do not respond to increases, although they fall slightly following declines.²¹ The transfer response therefore appears concentrated in programs tied to age rather than to economic need. This provides little

²¹Unemployment insurance and veterans' benefits do not respond in either direction. See Table D1 in Appendix D.

evidence that religious networks reduce reliance on public assistance through informal support provided by congregations.

Figure 8 suggests that changes in the adherent share move with the county's age structure. After an increase, the share of the population aged 65 and older rises by 0.25 percentage points at treatment and 0.46 points a decade later, against a sample mean of 14.9 percent, and median age rises by 0.23 and 0.63 years. After a decline, the retirement-age share falls by 0.31 points at treatment, though effects following a decrease in adherent share are generally insignificant. The 2000 cohort shows no pre-trend in either measure following increases. However, there is slight evidence of a pre-trend for median age following decreases, but the treatment effects are near zero in those cases regardless.

4.4 Labor Market Outcomes

If the increase in the adherent share correlates with an influx of older residents, we should see these effects in labor market outcomes as well. An older population has lower labor force participation, even if the number of local jobs remains unchanged. Figure 9 reports unemployment and labor force participation.

Following increases in the adherent share, labor force participation falls by 0.39 and 0.57 percentage points, and unemployment falls with a delay. Both patterns are consistent with an aging population, since retirees leave the labor force rather than become unemployed. Participation also falls after declines and unemployment rises. Here the decline cannot only reflect aging, since the retirement-age share falls after declines. Instead, it is consistent with the loss of jobs documented above, as some workers leave the labor force when local employment falls.

4.5 Separating the Demographic and Employment Channels

One concern is that the income response may be driven entirely by retirees moving into or out of these counties. The increase in jobs per person does not necessarily rule this out. Retiree arrivals could increase local demand for health care and other services, for exam-

ple, or lead to more part-time employment. To examine this possibility, we re-estimate our main specifications after removing counties that experience large contemporaneous changes in their retirement-age share, matching the exclusion to the direction of treatment. Specifically, when we estimate the effects of increases in the adherent share, we drop counties with a large *increase* in their retirement-age share. When we estimate the effects of declines, we drop counties with a large *decrease*. We define a large increase (decrease) as a decadal change in the retirement-age share more than one standard deviation above (below) the mean. For treated counties, we apply this rule to their treatment decade. For never-treated controls, we apply it to both 1980–1990 and 1990–2000. This removes 50 of 252 treated counties and 617 of 2,458 control counties from the analysis of increases, and 83 of 240 treated counties and 621 of 2,458 control counties from the analysis of declines. The results are reported in Figures A1 and A2 in Appendix A.

Excluding counties with large shifts in their retirement-age share leaves the effects on net income, net earnings, and jobs per person largely intact. After declines, net income, net earnings, and jobs per person fall. The decline in transfers, however, is much smaller and loses significance entirely for the 2000 cohort. After increases, net income and net earnings still rise with a delay, and jobs per person also increase. Social Security no longer responds, although total transfers and Medicare still rise. The retirement-age share also still rises after increases, but only in the 2000 cohort, which shows a significant pre-trend in this sample. Overall, the income and employment effects do not appear to depend on the counties with the largest shifts in age structure, while the decline in transfers after decreases in the adherent share largely does.

It is important to note that this exclusion exercise cannot tell us *why* the adherent share and age structure move together. We cannot fully distinguish between two possibilities. Causality may run from age to the adherent share. Older people tend to participate in religion at higher rates. As a result, the adherent share may increase simply because the county’s population gets older. This could occur as older households move into the county or as current residents age, without any underlying change in religious behavior.

Causality could also run in the other direction if an active religious community attracts older households deciding where to retire. Matching on pre-treatment age structure accounts for differences in levels, but not for changes during the treatment decade. The exclusion exercise shows that the employment results do not depend on counties with the largest contemporaneous shifts in age structure. It cannot rule out, however, that part of the income response reflects age-driven changes in the measured adherent share.

The employment results are somewhat different. Their persistence after excluding counties with the largest shifts in age structure makes demographic change, on its own, a less likely explanation, but leaves open what is driving them. One possibility is a network mechanism. Congregations are durable and spatially fixed with repeated interactions. This could lower search costs for both employers and workers. The time needed to build this kind of trust is also consistent with the delayed employment response after increases in adherence. Another possibility is local demand. Older residents spend their asset income and benefits locally, including on health care and other services, which could create jobs without raising pay per job. The exclusion exercise presented in this section makes this explanation less likely because the employment effects remain after dropping counties with the largest shifts in age structure. In either case, the absence of a change in pay per job or educational attainment points away from an individual-level mechanism in which religion raises worker productivity.

4.6 Robustness

In addition to the exclusion exercise in Section 4.5, Appendix A reports three robustness checks for each outcome. First, we use alternative treatment thresholds of 0.10 and 0.14. Second, we exclude “near-miss” controls, whose adherent share changed by more than 0.12 without meeting the persistence requirement. Third, for the 1990 cohort, we use the alternative treatment measure described in Section 2. Appendix B reports a fourth robustness check using the main treatment definitions with alternative estimators. Across these checks, the responses of net income, asset income, Social Security, and Medicare,

and the effects of declines on net earnings and jobs per person, are similar in sign and magnitude. Two results are slightly weaker. The rise in jobs per person after increases is smaller under the alternative thresholds. In addition, the pooled effect of increases on unemployment a decade after treatment, which is only marginally significant in the main specification, falls just short of significance when near-miss controls are excluded, although the point estimate is nearly unchanged. Under the alternative treatment measure, the demographic responses are, if anything, stronger, although the earnings estimates are imprecise given the small number of treated counties. Thus, our results do not seem to depend on the exact treatment definition, control group composition, or estimator.

5 Conclusion

This paper shows that large, sustained changes in the adherent share are followed by meaningful changes in county-level income per capita. Counties with large declines in the adherent share see net income per capita fall by about four percent at the time of the decline and seven percent a decade later. Likewise, counties with large increases see gains that emerge only with a delay, of five to seven percent within one to two decades.

Decomposing income reveals two channels. Changes in the adherent share move with the local retirement-age population, and much of the immediate income response comes through asset income rather than through earnings. Age-related transfers (Social Security and Medicare) also increase. Separately, jobs per person and earnings rise with the adherent share and fall with its decline, with no change in pay per job. Importantly, this employment effect survives excluding counties with simultaneous large shifts in their retirement-age population.

These results help reconcile the positive income effects we estimate with the weak or negative relationships between religion and county output reported in prior work. Religious communities appear to shape local economies less by raising the productivity of individual workers and more by shaping who lives in a place and how many of them

are employed. Measures of local economic performance that count only production, or only earnings, will not fully capture this.

Our findings carry two implications for local economic policy. First, large declines in religious congregations are followed by measurable economic losses. Counties with large, sustained declines in the adherent share lose jobs and earnings relative to comparable counties, and the loss is larger (but more imprecise) once shifts in the retirement-age population are set aside. Second, part of the income associated with growing religious communities reflects an older population. That part of the income response comes from asset income rather than earnings. Social Security and Medicare receipts also increase, leaving total personal income in these counties more closely tied to federal programs and to changes in the age structure of the population.

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Tables and Figures

Table 1: Sample Composition by Treatment Period

Treatment Period	Treated	Treatment Year	Covariate Year	Outcome Years
<i>Positive Change in Religious Adherents</i>				
1980–1990	147	1990	1980	1990, 2000, 2010
1990–2000	105	2000	1990	2000, 2010
<i>Negative Change in Religious Adherents</i>				
1980–1990	53	1990	1980	1990, 2000, 2010
1990–2000	187	2000	1990	2000, 2010
Total Positive Treatments			252	
Total Negative Treatments			240	
Control (Never Treated)			2,458 per stack	

Notes: Treatment is defined as a sustained jump in religious adherents between periods. Control counties are *never treated* in that they do not experience a large increase or decrease in the share of adherents within our sample and appear in each stack. Covariates are lagged 10 years prior to the treatment year for matching purposes.

Table 2: Baseline Summary Statistics

	Mean	SD
<i>Main outcome: Productive income per-capita</i>		
Net income per capita (logged)	10.072	0.283
<i>Key matching covariates (lagged by 10)</i>		
Net income per capita (logged)	9.868	0.282
Adherent share	0.567	0.183
Young population %	0.230	0.030
Teen population %	0.100	0.022
Working age population %	0.528	0.035
Male population %	0.491	0.016
Urban Indicator	0.323	0.468
High school % of adults	0.652	0.121
College Degree % of adults	0.125	0.059
Military % of employment	0.008	0.039
Unemployment rate	0.068	0.032
Labor force participation rate	0.595	0.067
<i>Secondary outcomes</i>		
Net earnings per capita (logged)	9.787	0.305
Dividends, interest, and rent per capita (logged)	8.637	0.347
Jobs per person	0.504	0.141
Wage per job (logged)	10.354	0.212
Retirement-age population share	0.149	0.042
Median age (years)	35.905	4.016
Government transfers per capita (logged)	8.405	0.290
Social Security per capita (logged)	7.592	0.291
Medicare per capita (logged)	6.757	0.388
Medicaid per capita (logged)	6.411	0.661
Income maintenance per capita (logged)	5.965	0.529

N=5,168 potential county-stack observations. Secondary outcomes are measured in the baseline year. This full sample includes all positive treatments and never treated controls. This table is replicated for negative treatments in Appendix C. All covariates are lagged 10 years prior to the event for each stack. The matching covariates also include net income per capita lagged 20 years, which is not shown. For jumps in the adherent share between 1980 and 1990, 1980 values are used for covariate matching. For 1990-2000 treatments, 1990 values are used.

Table 3: Unweighted Difference-in-Means in T -10: Treated (Positive Jumps) vs Control (Never Treated)

	Control	Treated	Difference	P-value
<i>Matching covariates</i>				
Net income per capita (logged)	9.867	9.881	0.014	0.454
Adherent share	0.567	0.566	-0.001	0.951
Young population %	0.230	0.232	0.002	0.394
Teen population %	0.100	0.098	-0.002	0.104
Working age population %	0.529	0.520	-0.009	0.000
Male population %	0.491	0.492	0.002	0.101
Urban indicator	0.325	0.290	-0.035	0.248
High school % of adults	0.652	0.656	0.005	0.547
College degree % of adults	0.124	0.130	0.006	0.121
Military % of employment	0.008	0.009	0.001	0.718
Unemployment rate	0.068	0.052	-0.016	0.000
Labor force participation rate	0.595	0.595	0.000	0.971
<i>Secondary outcomes (lagged by 10)</i>				
Net earnings per capita (logged)	9.580	9.576	-0.004	0.860
Dividends, interest, and rent per capita (logged)	8.427	8.514	0.087	0.000
Jobs per person	0.462	0.494	0.032	0.000
Wage per job (logged)	10.260	10.239	-0.021	0.100
Retirement-age population share	0.141	0.151	0.010	0.000
Median age (years)	32.744	33.316	0.573	0.031
Government transfers per capita (logged)	8.101	8.050	-0.050	0.008
Social Security per capita (logged)	7.351	7.341	-0.010	0.642
Medicare per capita (logged)	6.215	6.218	0.003	0.910
Medicaid per capita (logged)	5.654	5.521	-0.133	0.005
Income maintenance per capita (logged)	5.753	5.546	-0.207	0.000

Treated $N = 252$, Control $N = 4,916$. Difference is treated minus control. Secondary outcomes enter the matching set only in their own specifications, through their own lags. P -values from two-sided t-tests. Summary statistics and differences-in-means tests for negative treatments are reported in Appendix C.

Figure 1: Share of adherents through time for treatments (positive and negative).

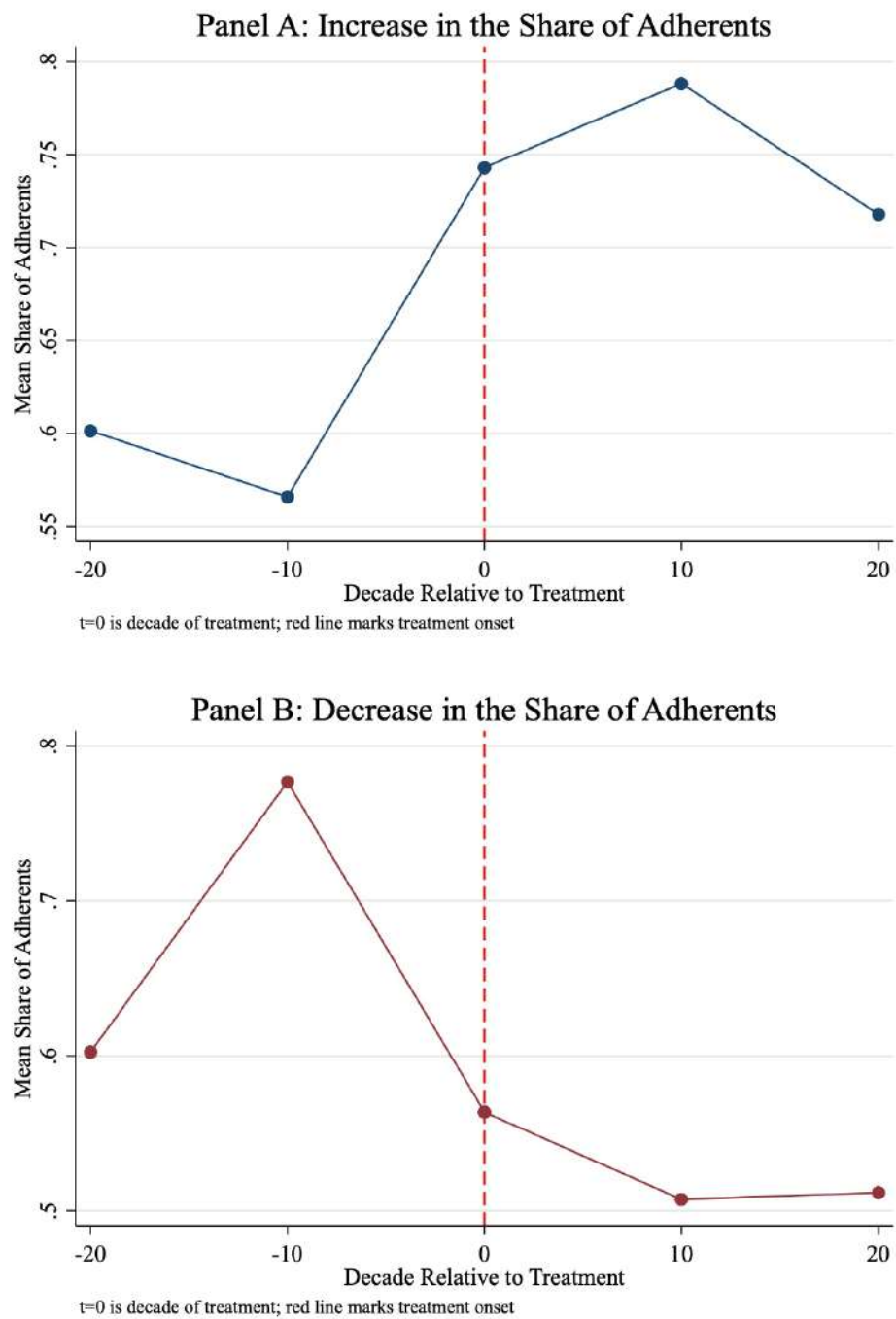
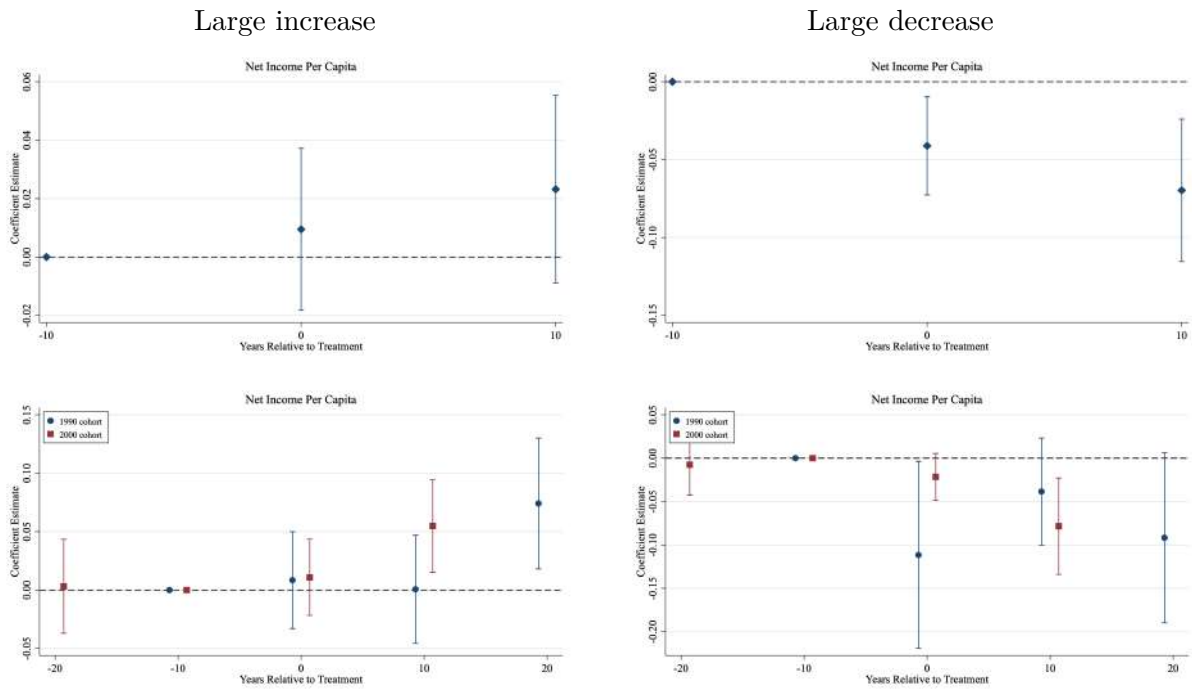


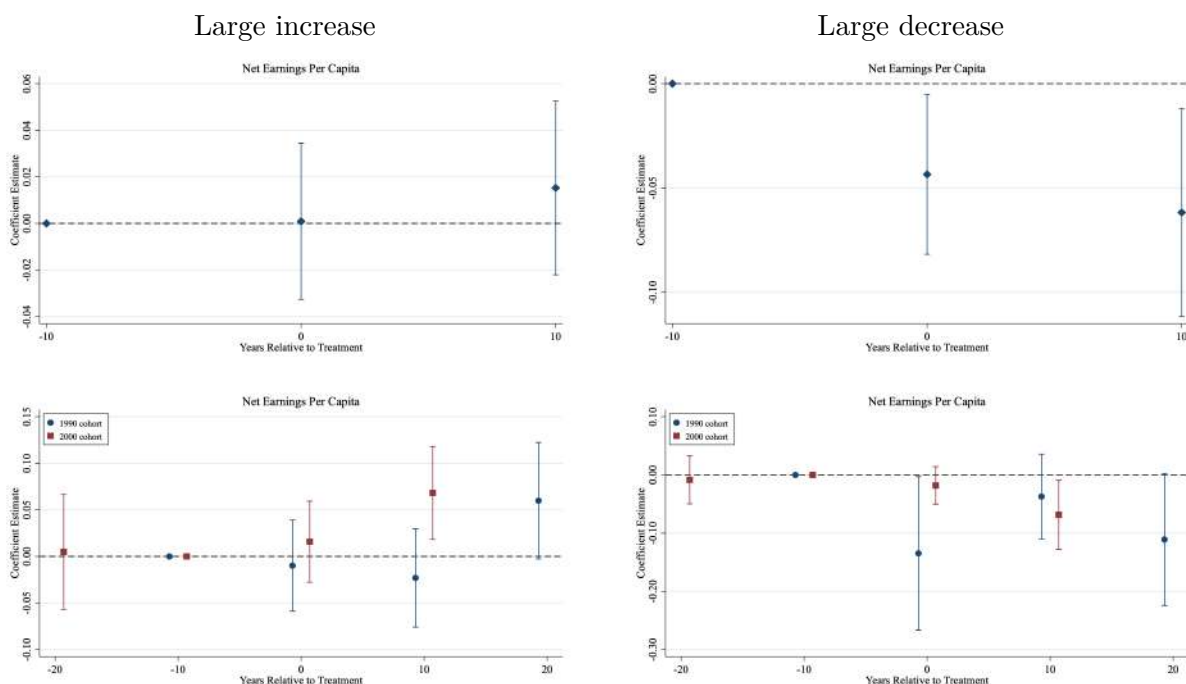
Figure 2: The effect of large changes in the adherent share on net income per capita



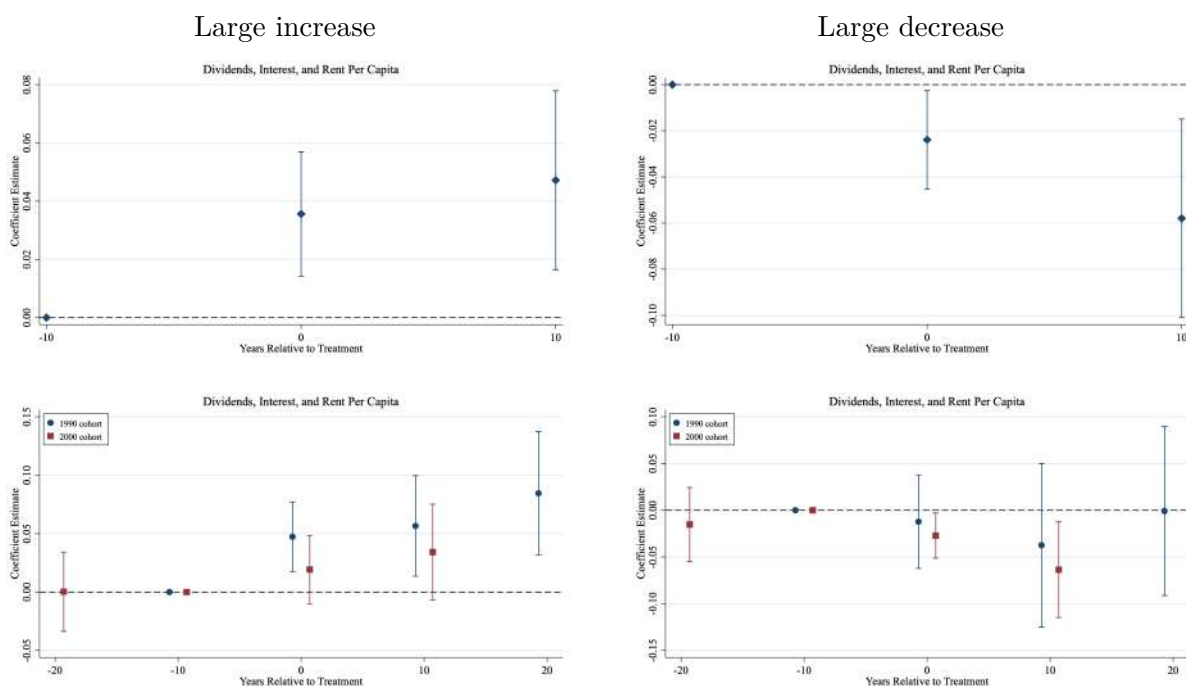
Notes: 95% confidence intervals. Stacked difference-in-differences estimates with normalized inverse propensity score weights; $k = -10$ is the reference period. Matching covariates are those in Table 3 plus income at $T - 20$. Standard errors clustered by county.

Figure 3: The components of net income: net earnings and dividends, interest, and rent

Panel A: Net Earnings Per Capita



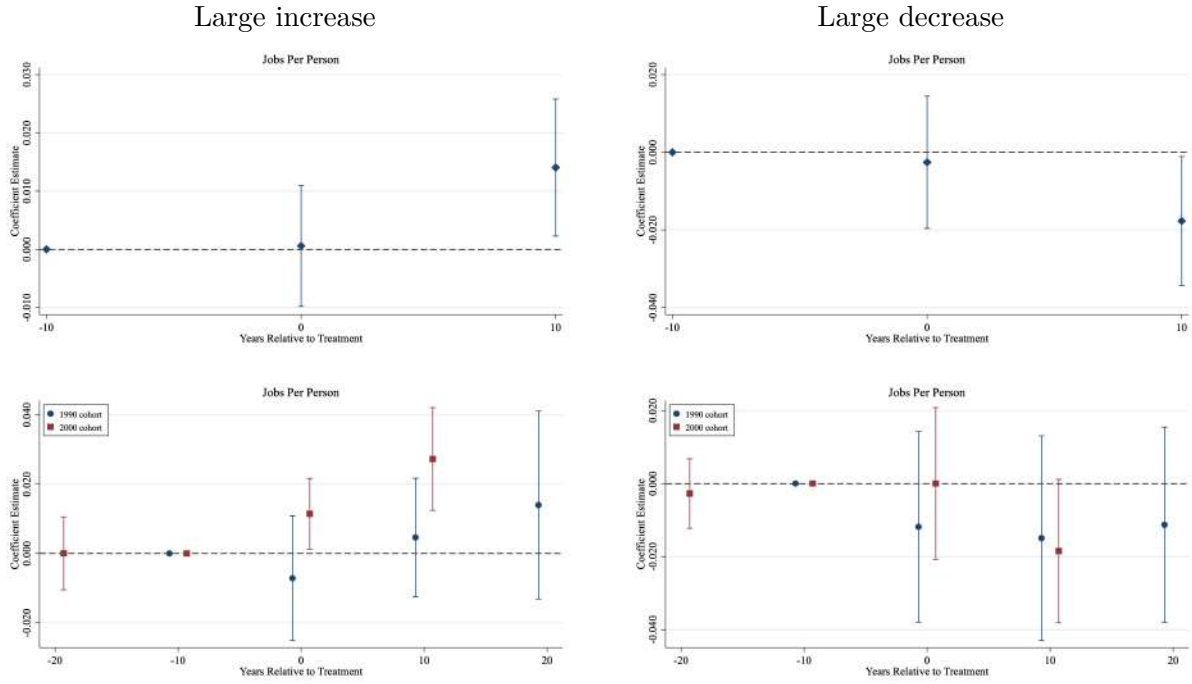
Panel B: Dividends, Interest, and Rent Per Capita



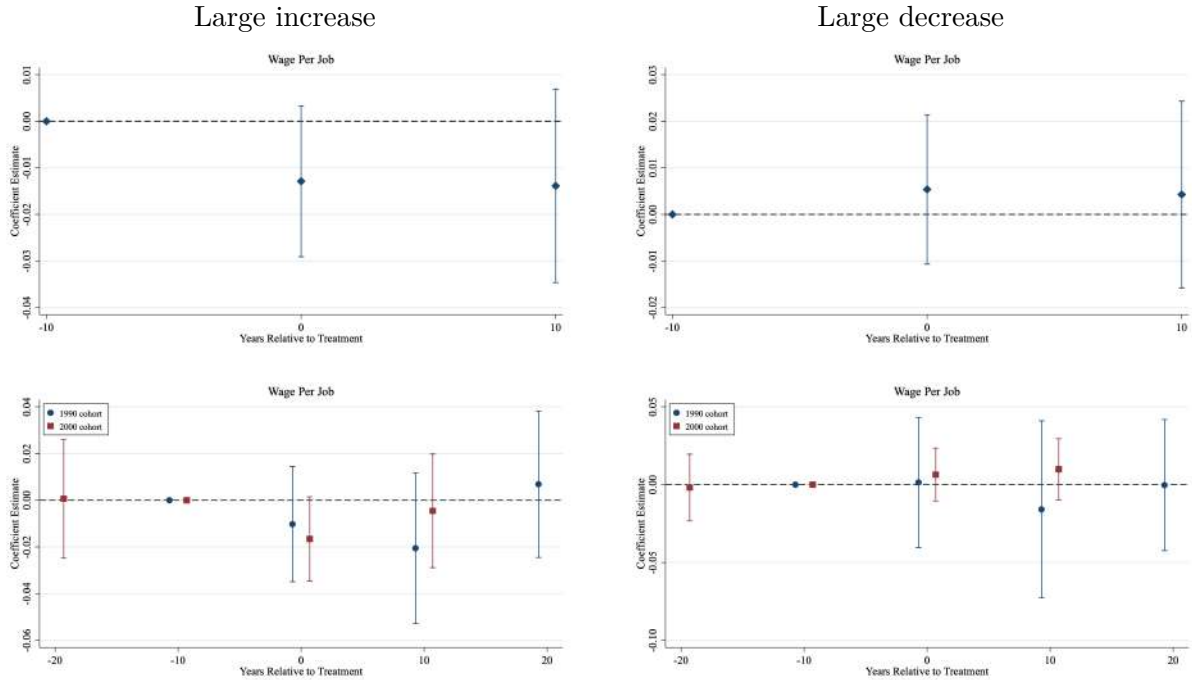
Notes: 95% confidence intervals. Stacked difference-in-differences estimates with normalized inverse propensity score weights; $k = -10$ is the reference period. Matching covariates are those in Table 3 plus income at $T - 20$, and two lags of the outcome. Standard errors clustered by county. Outcomes are in logs.

Figure 4: Employment and pay: jobs per person and wage per job

Panel A: Jobs Per Person

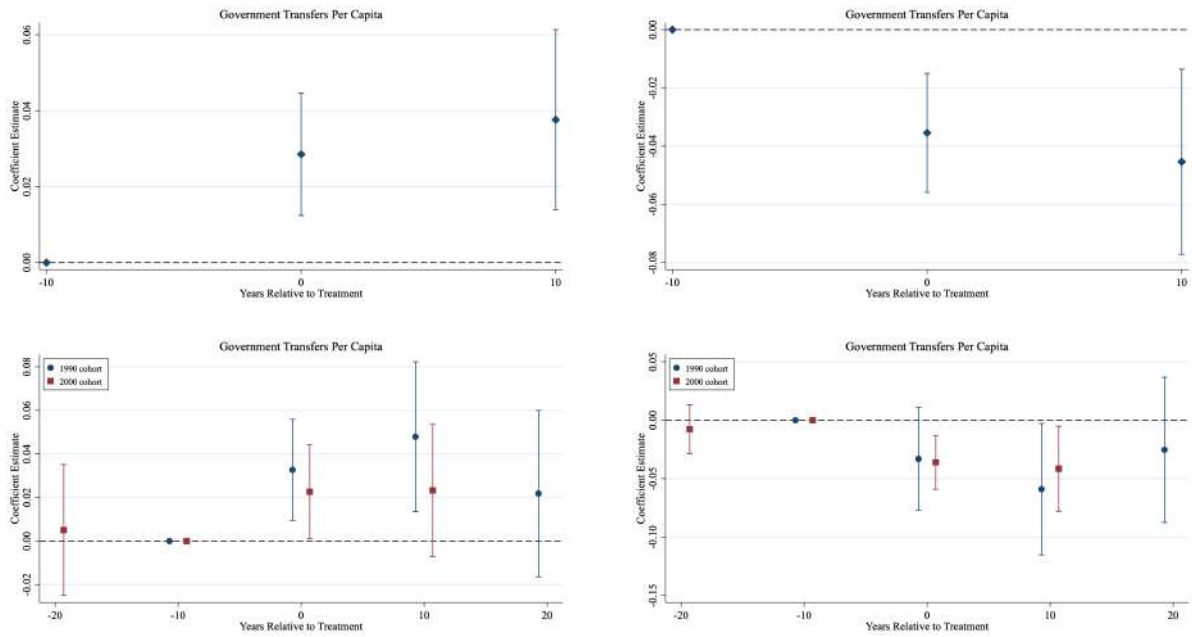


Panel B: Wage Per Job



Notes: 95% confidence intervals. Stacked difference-in-differences estimates with normalized inverse propensity score weights; $k = -10$ is the reference period. Matching covariates are those in Table 3 plus income at $T - 20$, and two lags of the outcome. Standard errors clustered by county. Wage per job is in logs.

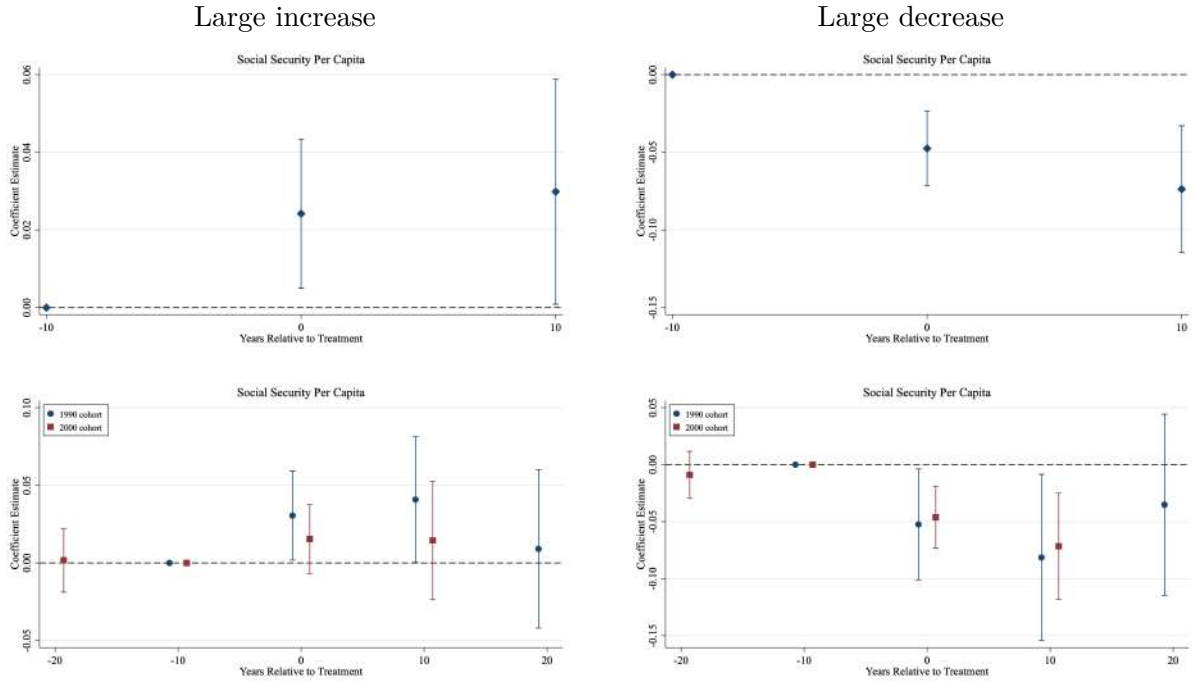
Figure 5: Total government transfers per capita



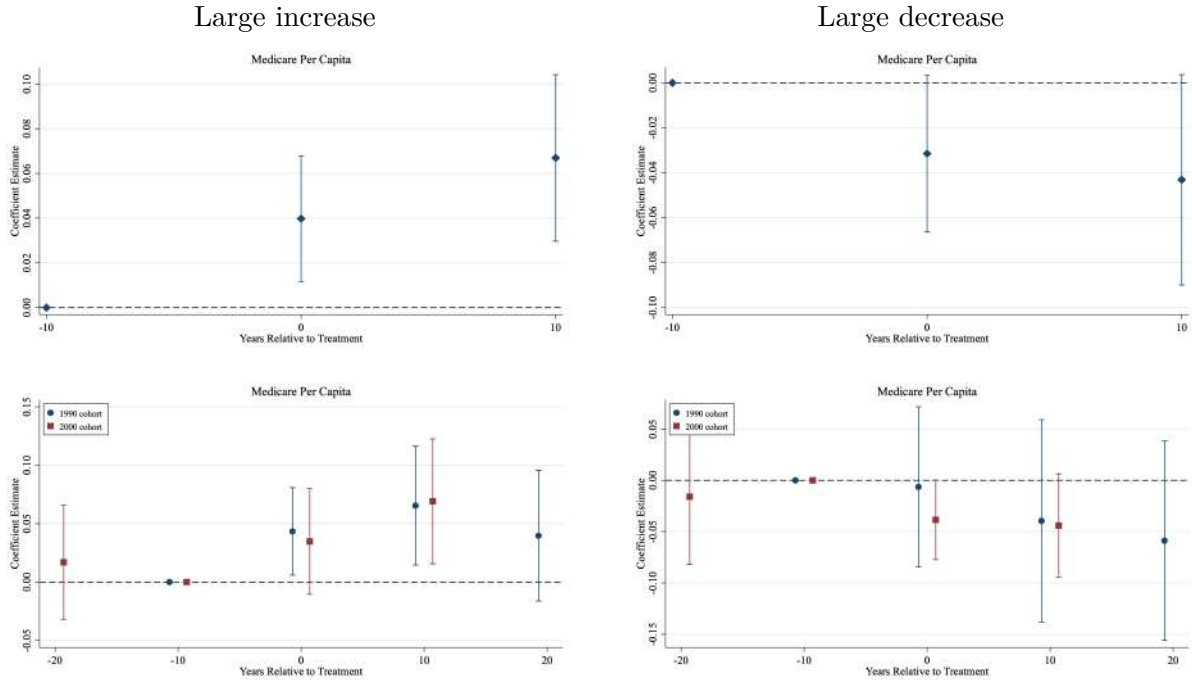
Notes: 95% confidence intervals. Stacked difference-in-differences estimates with normalized inverse propensity score weights; $k = -10$ is the reference period. Matching covariates are those in Table 3 plus income at $T - 20$, and two lags of the outcome. Standard errors clustered by county. Outcome is in logs.

Figure 6: Government transfers by program: Social Security and Medicare

Panel A: Social Security Per Capita



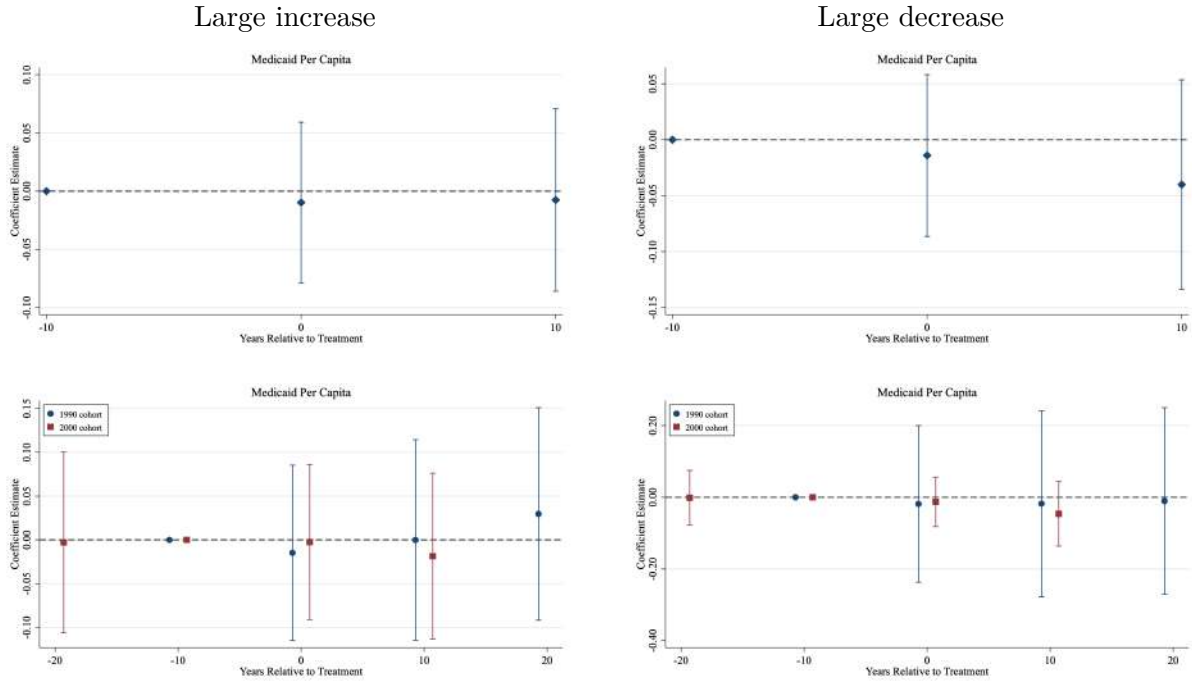
Panel B: Medicare Per Capita



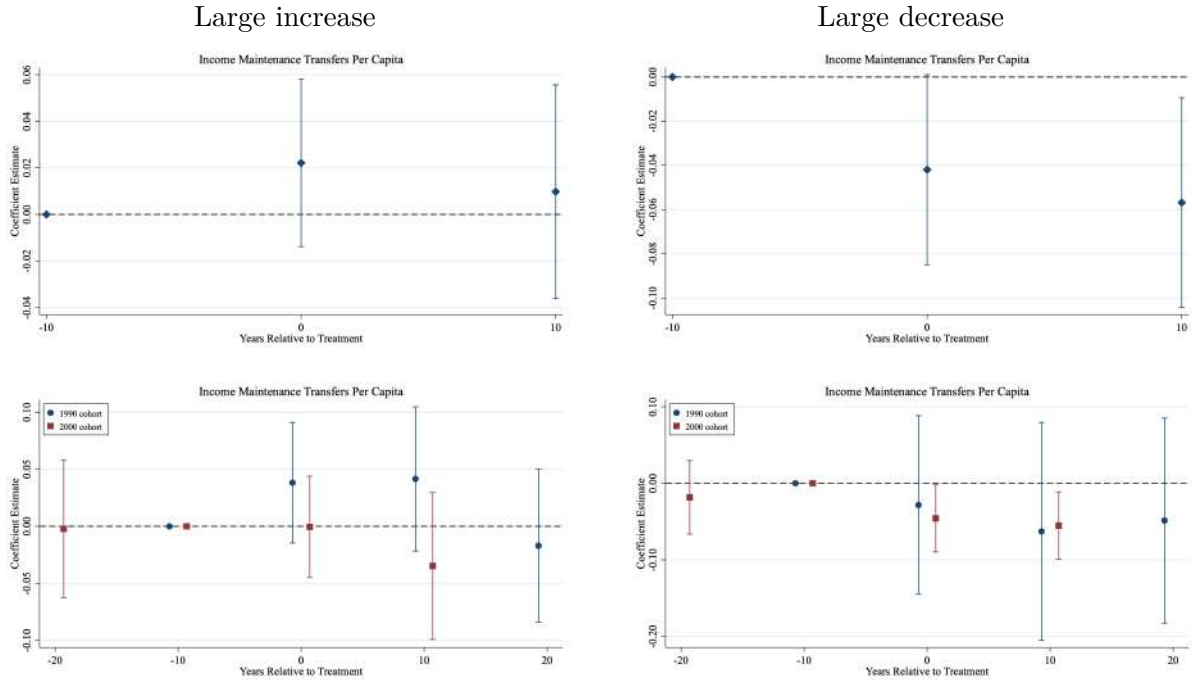
Notes: 95% confidence intervals. Stacked difference-in-differences estimates with normalized inverse propensity score weights; $k = -10$ is the reference period. Matching covariates are those in Table 3 plus income at $T - 20$, and two lags of the outcome. Standard errors clustered by county. Outcomes are in logs.

Figure 7: Government transfers by program: Medicaid and income maintenance

Panel A: Medicaid Per Capita



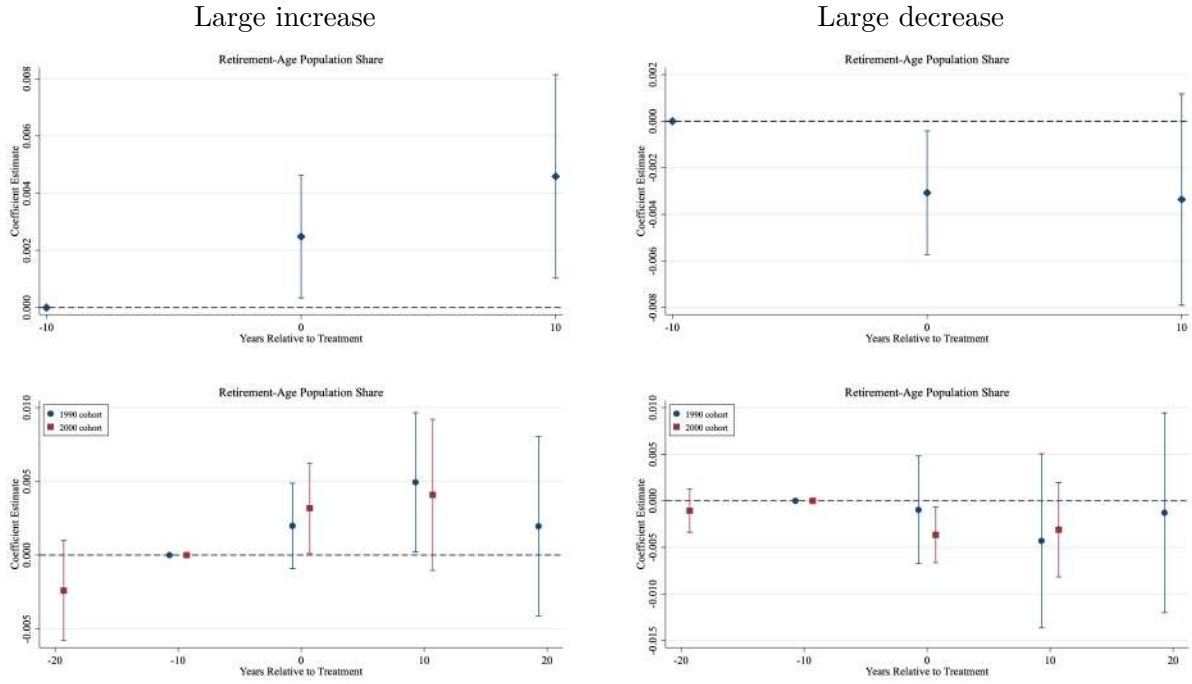
Panel B: Income Maintenance Transfers Per Capita



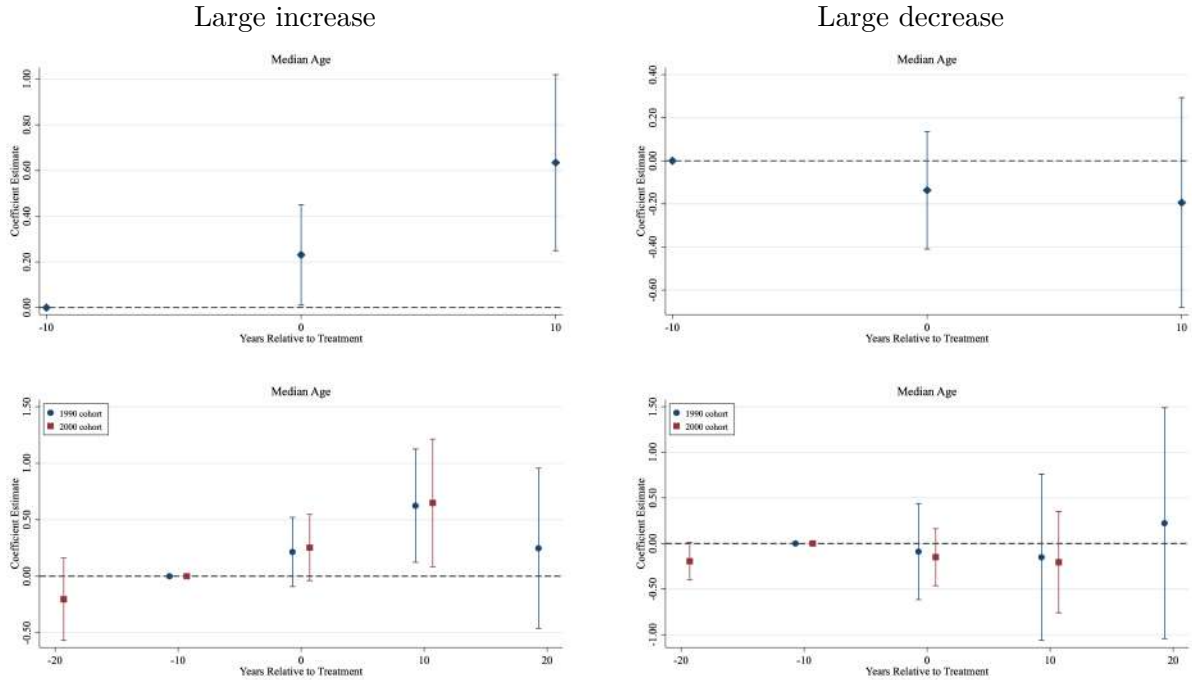
Notes: 95% confidence intervals. Stacked difference-in-differences estimates with normalized inverse propensity score weights; $k = -10$ is the reference period. Matching covariates are those in Table 3 plus income at $T - 20$, and two lags of the outcome. Standard errors clustered by county. Outcomes are in logs; Medicaid is BEA public assistance medical care.

Figure 8: Demographic composition: retirement-age share and median age

Panel A: Retirement-Age Population Share



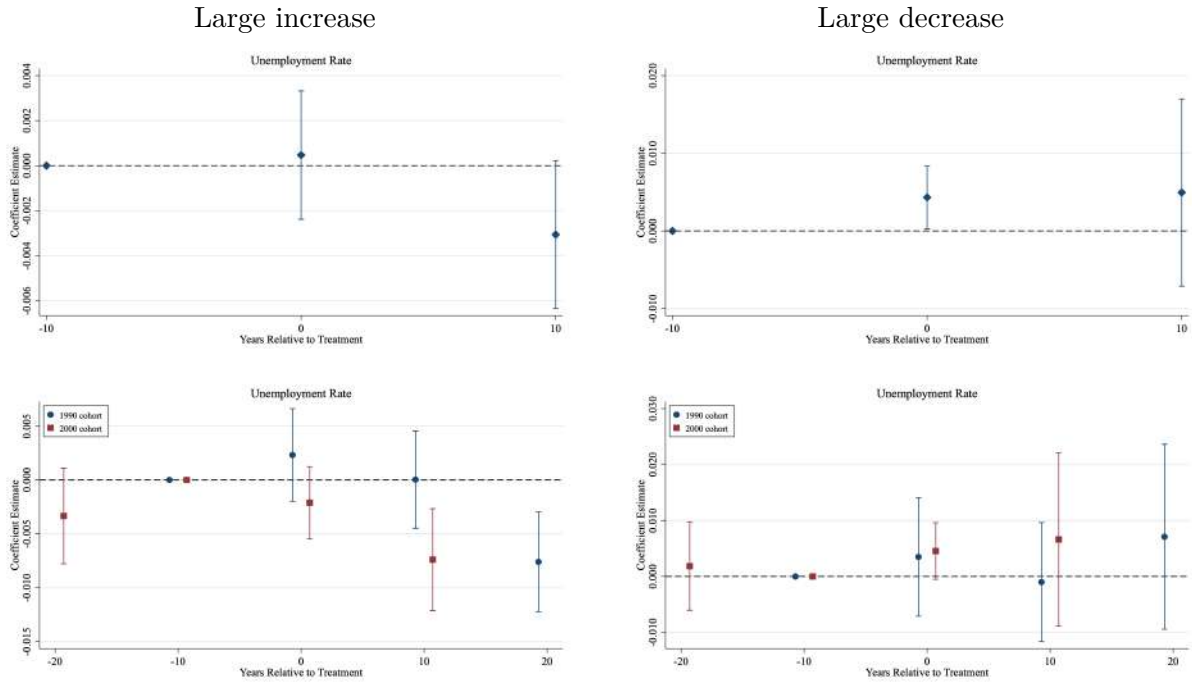
Panel B: Median Age



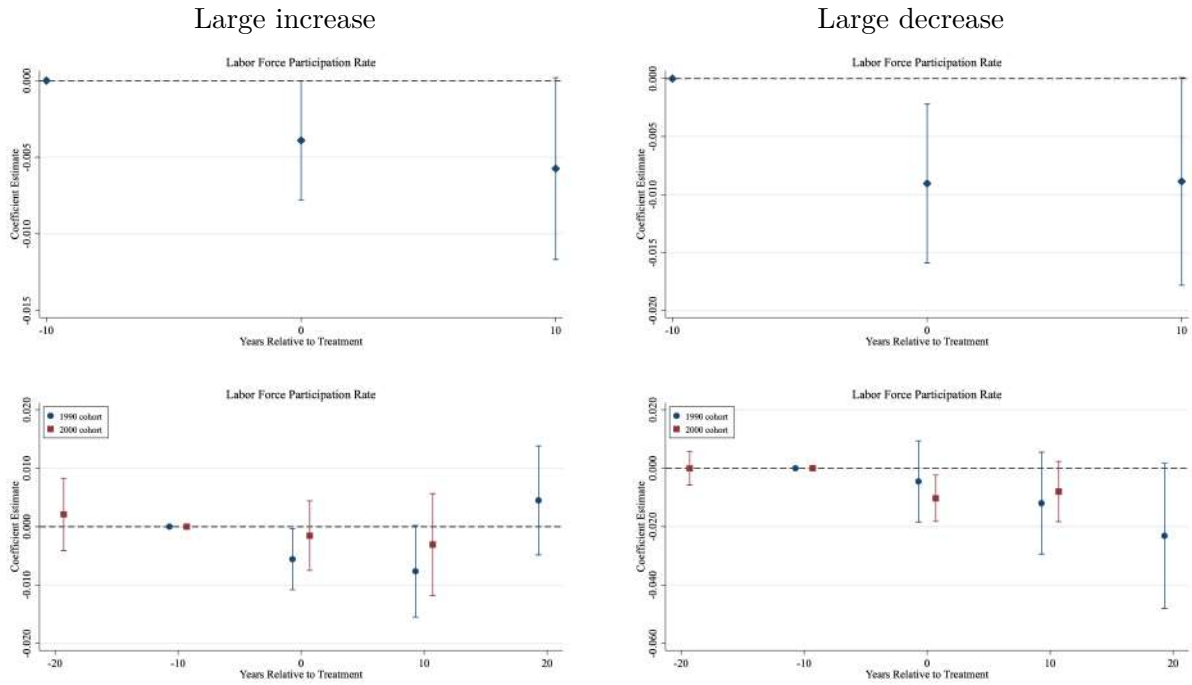
Notes: 95% confidence intervals. Stacked difference-in-differences estimates with normalized inverse propensity score weights; $k = -10$ is the reference period. Matching covariates are those in Table 3 plus income at $T - 20$, and the ten-year lag of the outcome. Standard errors clustered by county. Retirement age is 65 and older.

Figure 9: Labor market outcomes: unemployment and labor force participation

Panel A: Unemployment Rate



Panel B: Labor Force Participation Rate



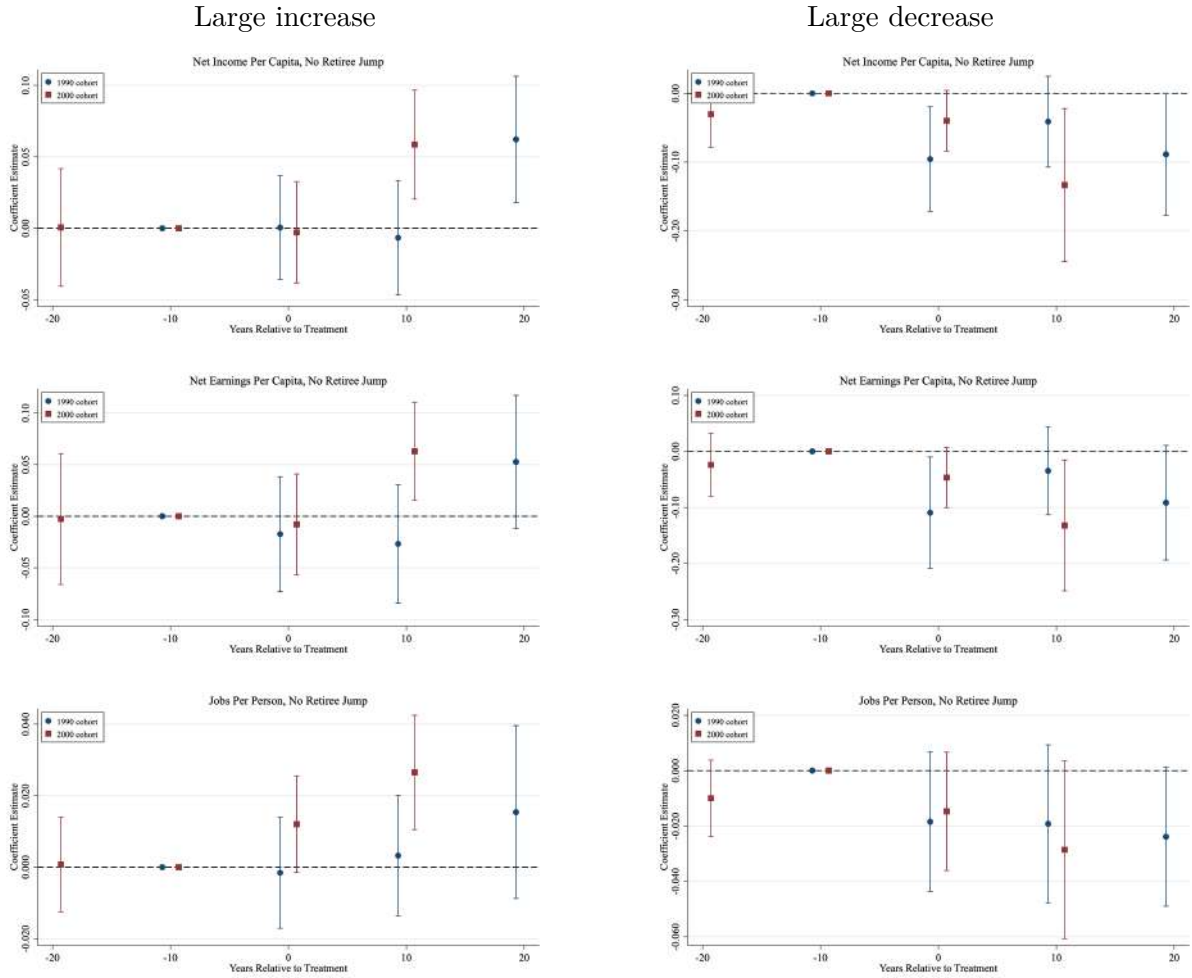
Notes: 95% confidence intervals. Stacked difference-in-differences estimates with normalized inverse propensity score weights; $k = -10$ is the reference period. Matching covariates are those in Table 3 plus income at $T - 20$. Standard errors clustered by county.

Appendix

A Robustness

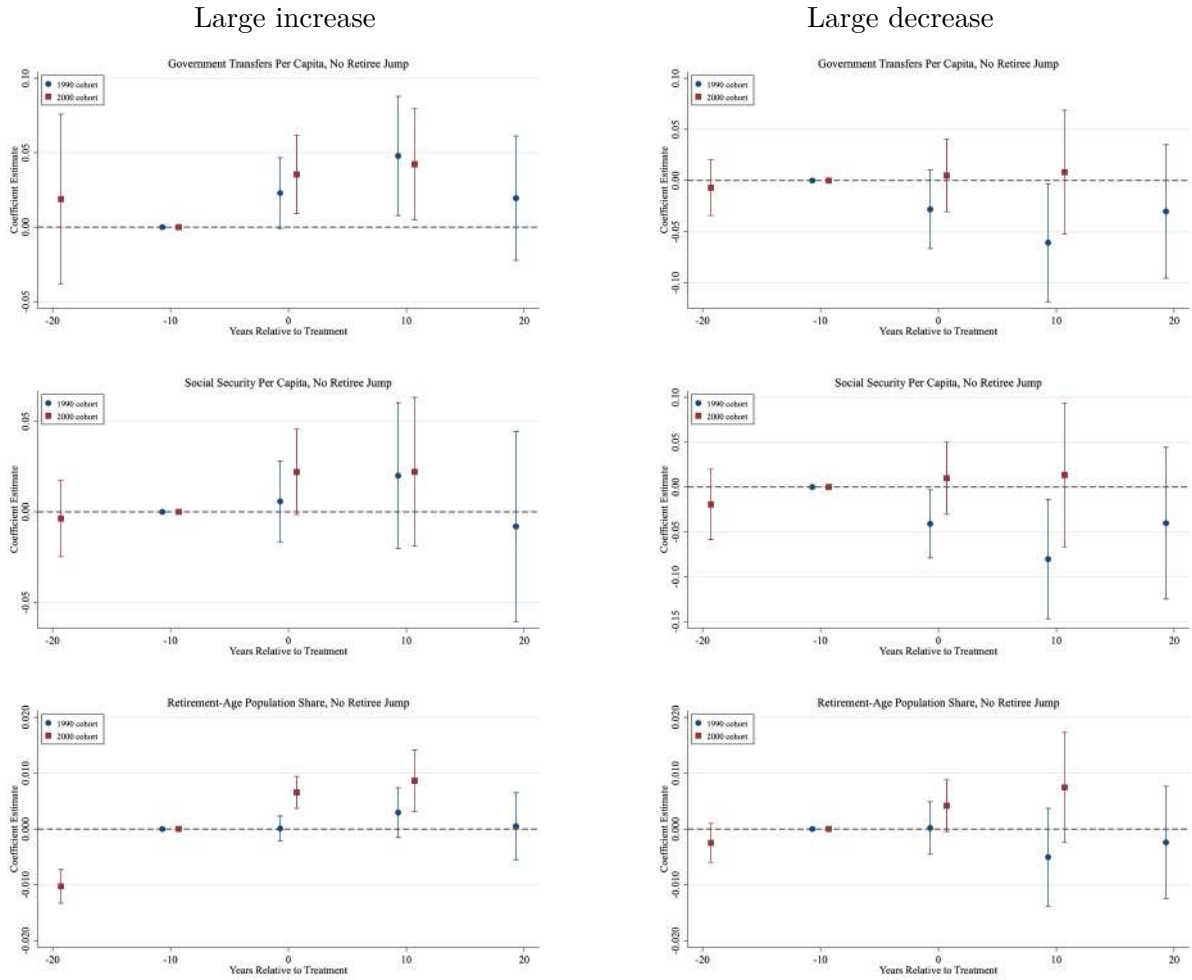
Excluding counties with contemporaneous large shifts in the retirement-age population

Figure A1: Net income, earnings, and employment, excluding counties with sharp shifts in the retirement-age share



Notes: 95% confidence intervals. Rows: log net income per capita, log net earnings per capita, jobs per person. Excludes treated counties whose retirement-age share changed by more than one standard deviation above (left) or below (right) the mean in their treatment decade, and controls with such a change in 1980–1990 or 1990–2000. Cohort-specific estimates; specifications otherwise match the main text. Standard errors clustered by county.

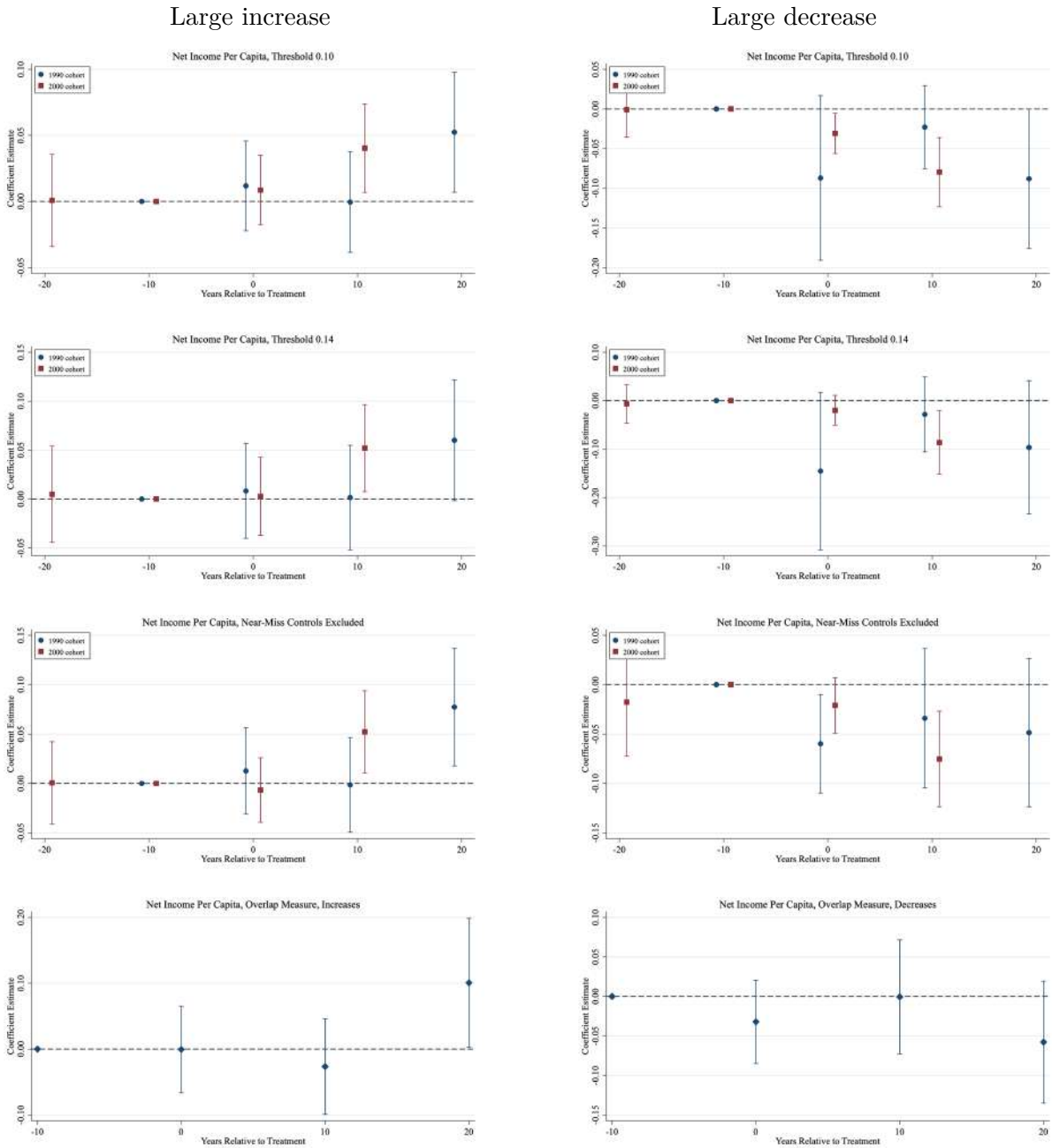
Figure A2: Transfers and the retirement-age share, excluding counties with sharp shifts in the retirement-age share



Notes: 95% confidence intervals. Rows: log government transfers per capita, log Social Security per capita, retirement-age population share. Excludes treated counties whose retirement-age share changed by more than one standard deviation above (left) or below (right) the mean in their treatment decade, and controls with such a change in 1980–1990 or 1990–2000. Cohort-specific estimates; specifications otherwise match the main text. Standard errors clustered by county.

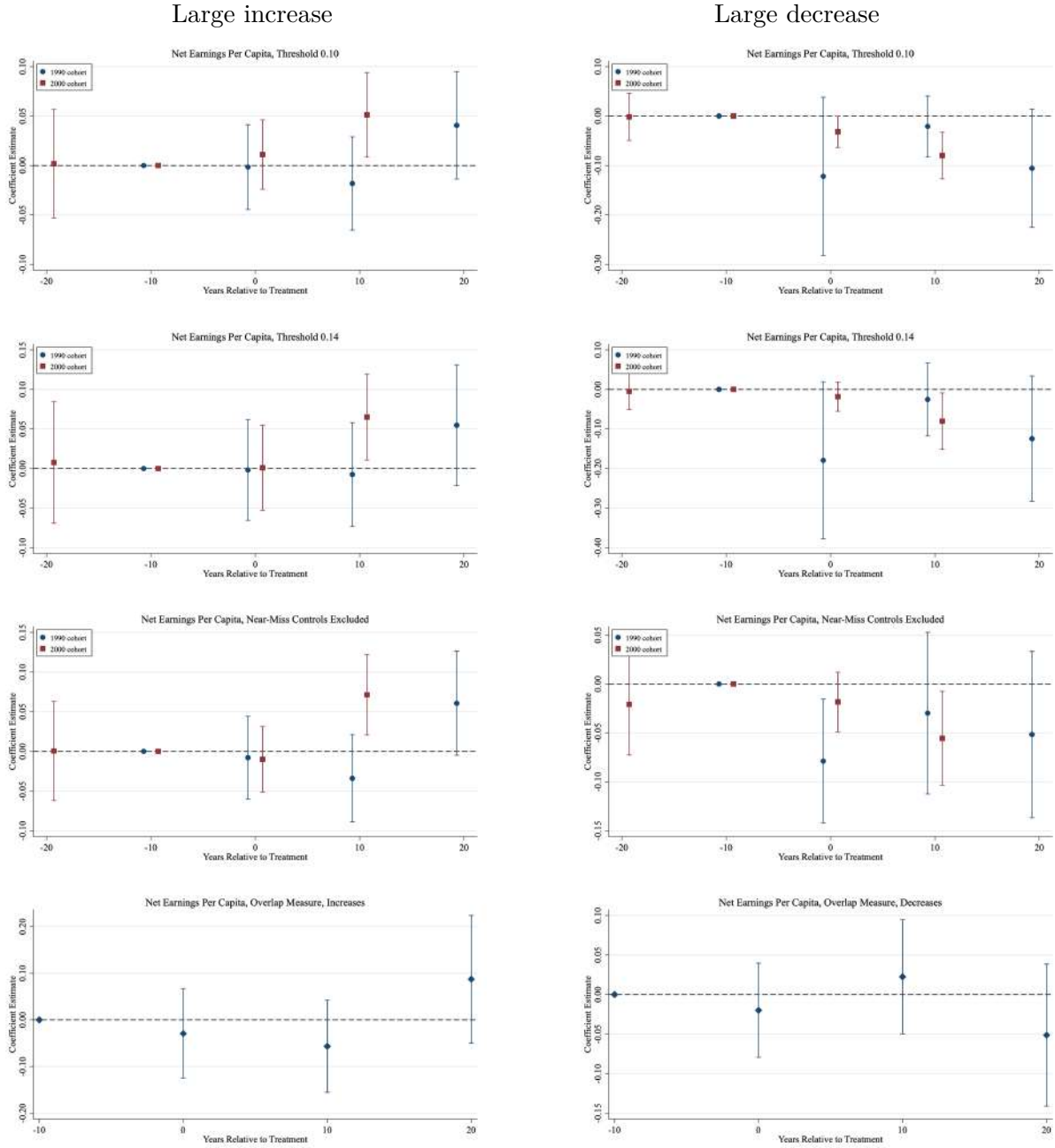
Alternative treatment definitions

Figure A3: Robustness to alternative treatment definitions: log net income per capita



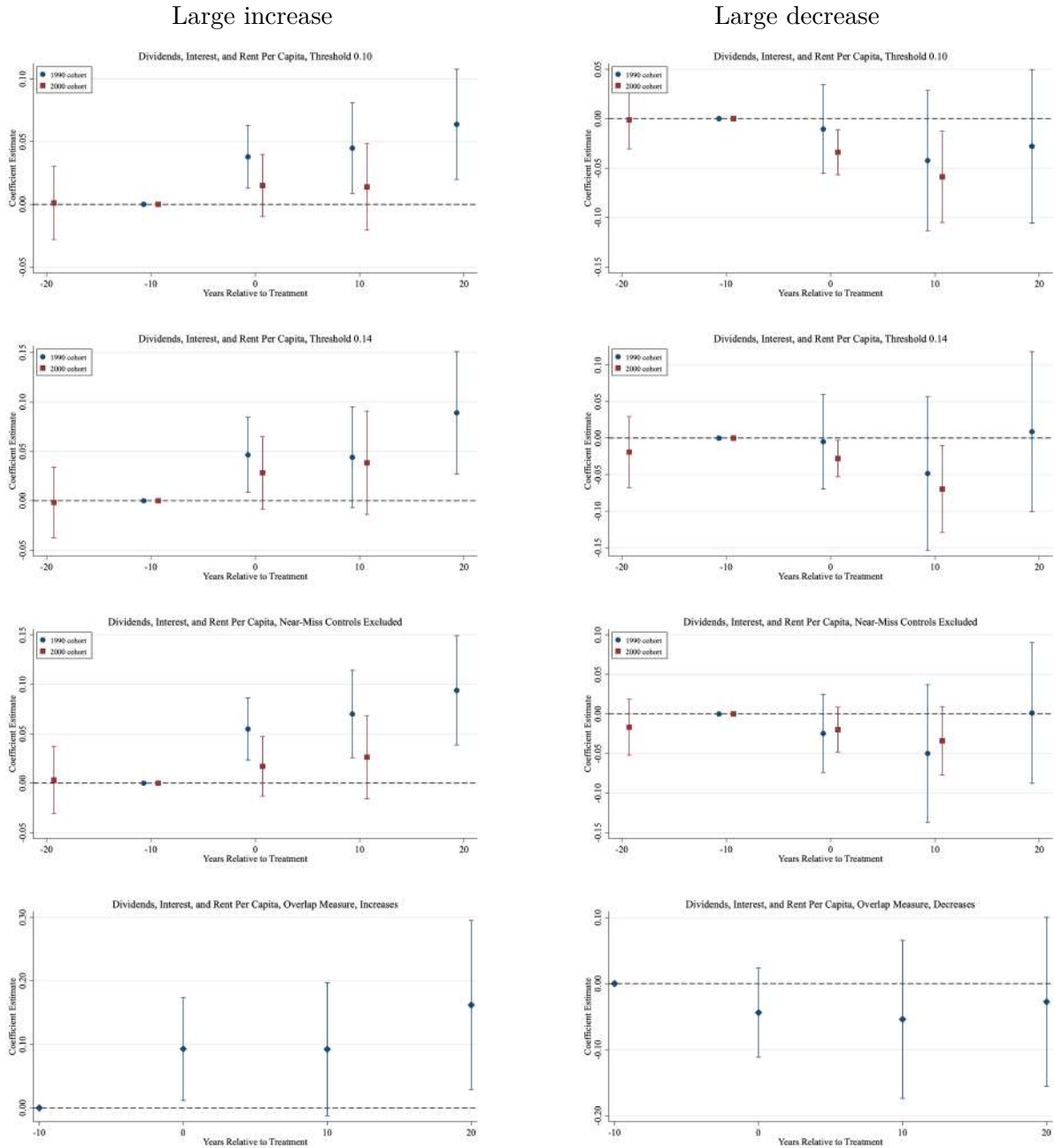
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A4: Robustness to alternative treatment definitions: log net earnings per capita



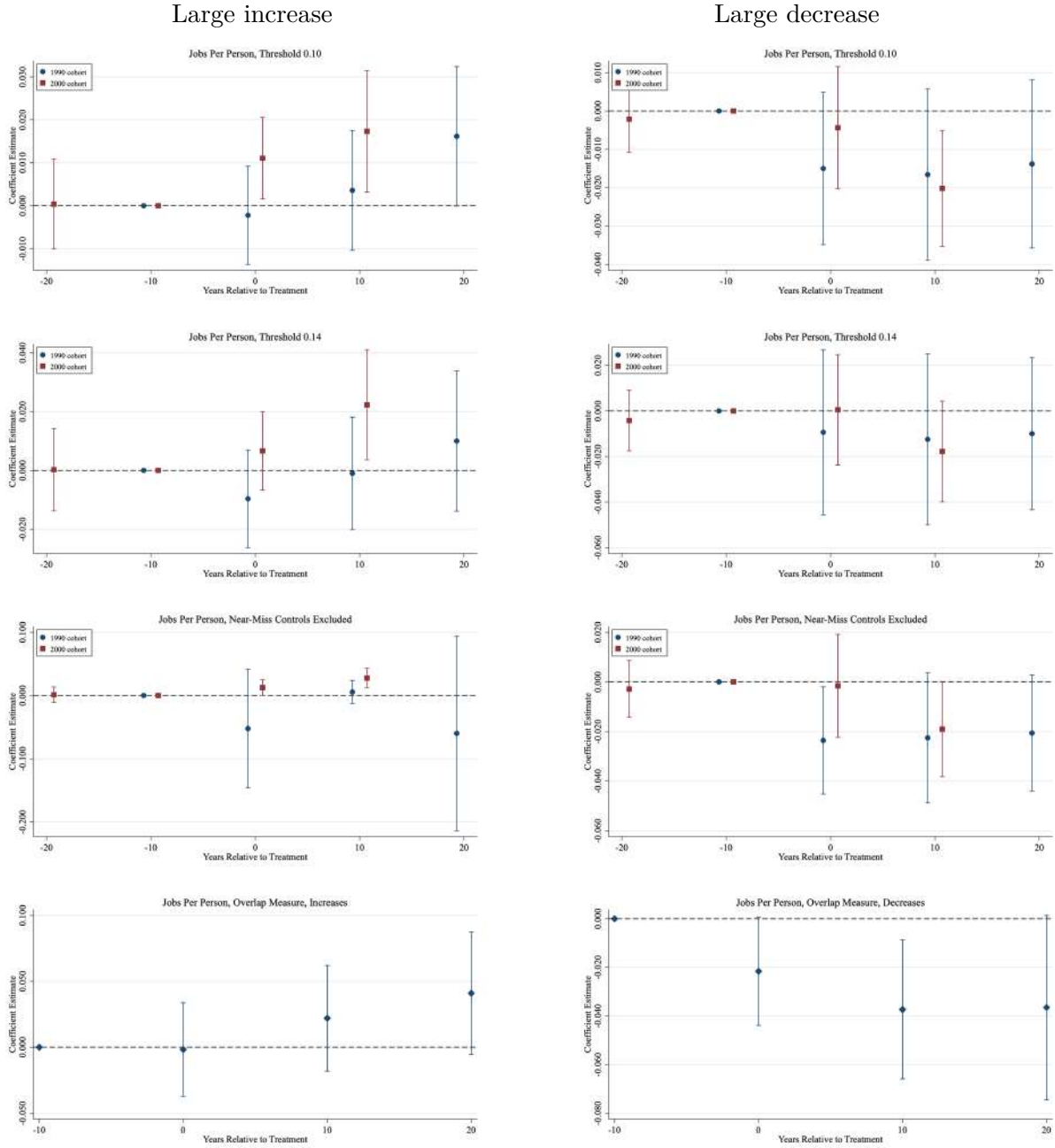
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A5: Robustness to alternative treatment definitions: log dividends, interest, and rent per capita



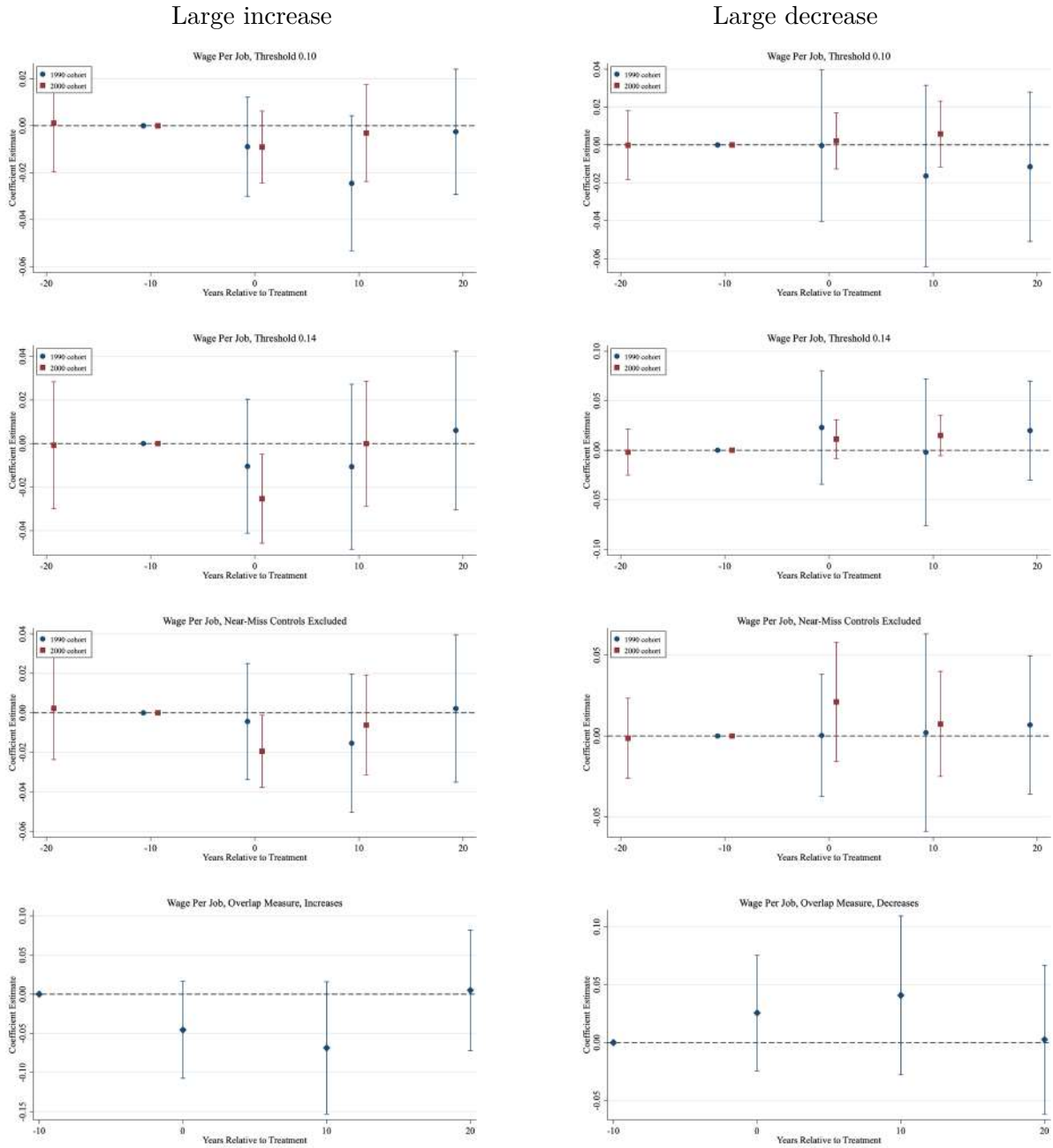
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A6: Robustness to alternative treatment definitions: jobs per person



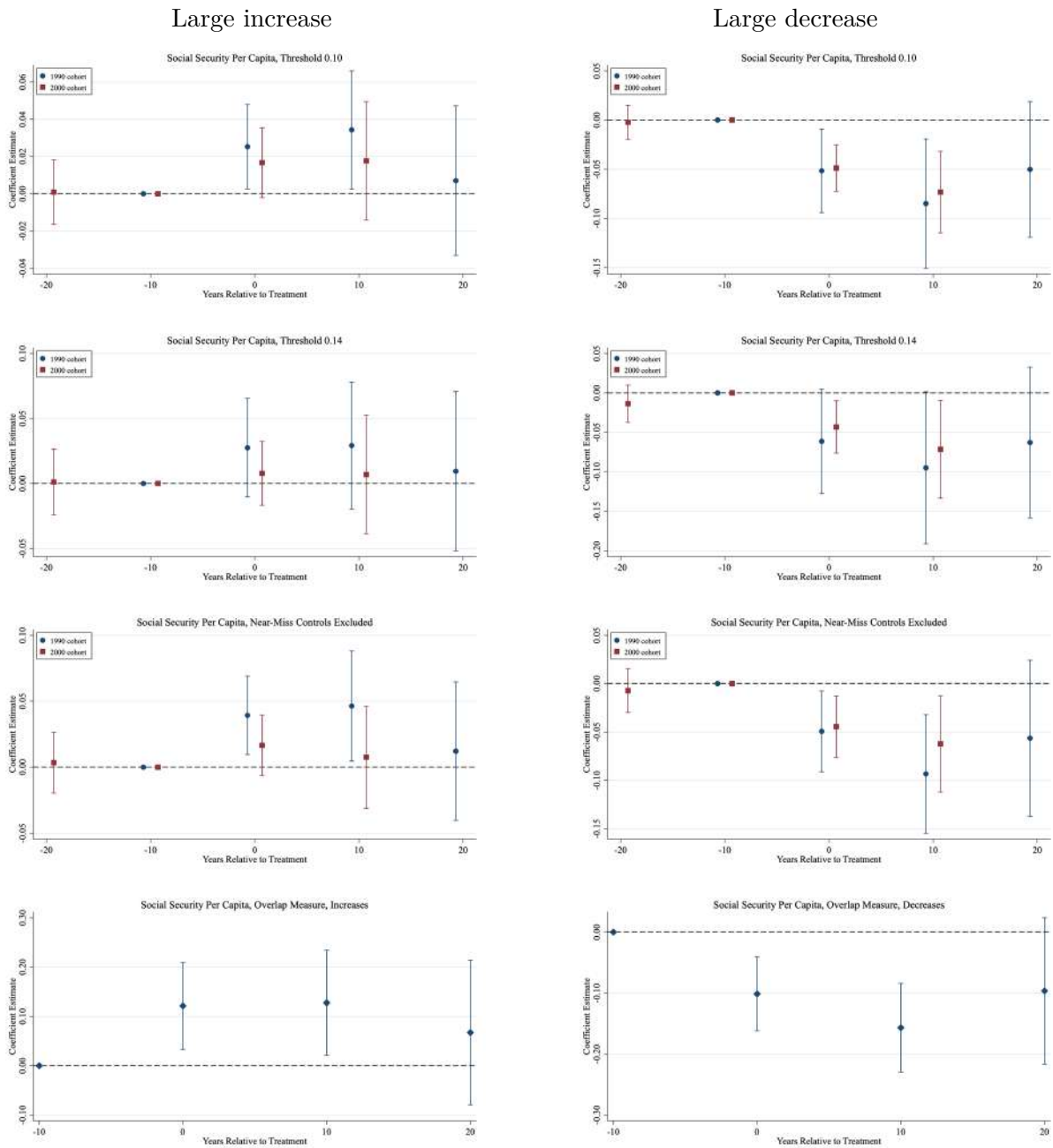
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A7: Robustness to alternative treatment definitions: log wage per wage and salary job



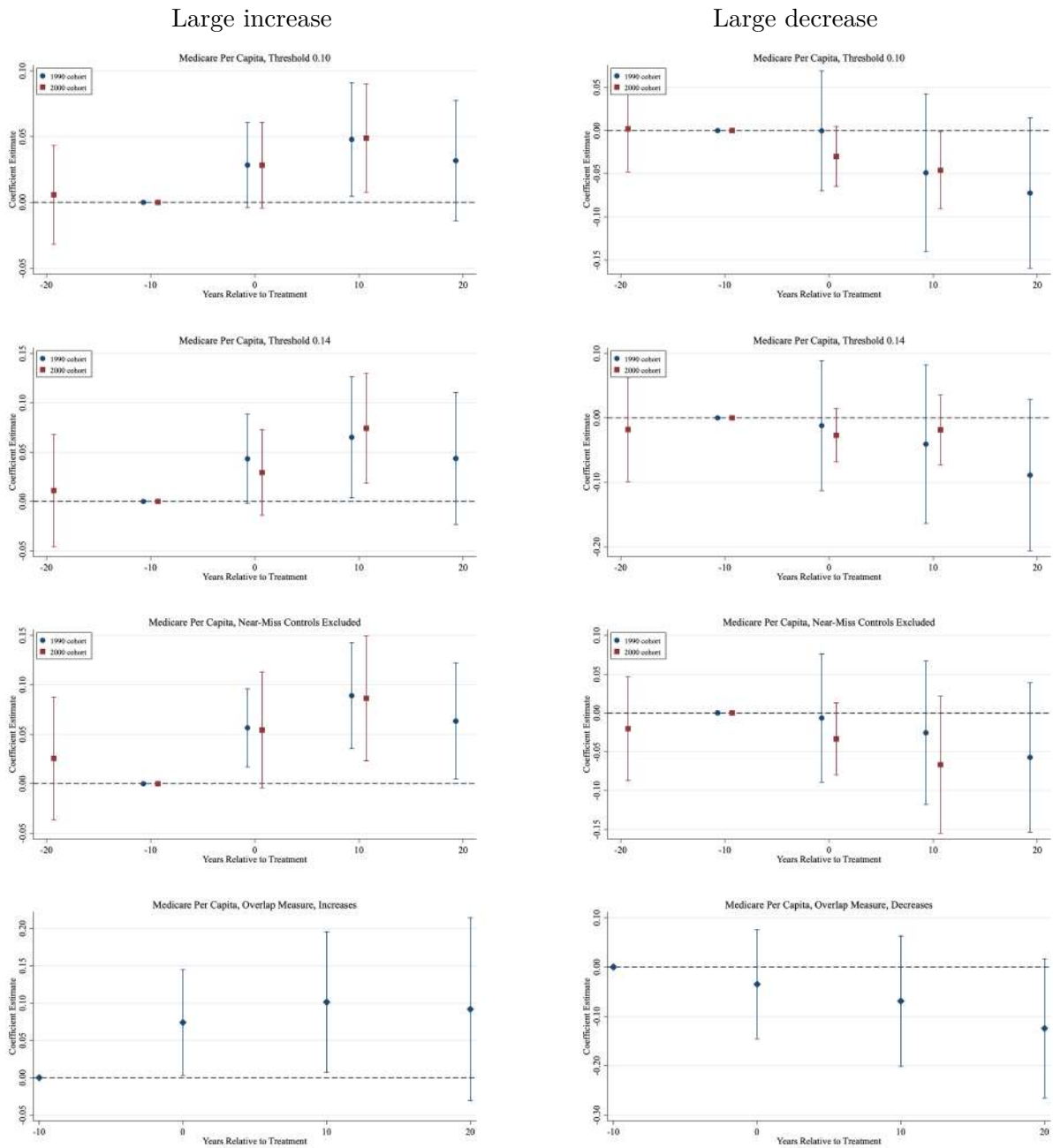
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A8: Robustness to alternative treatment definitions: log Social Security per capita



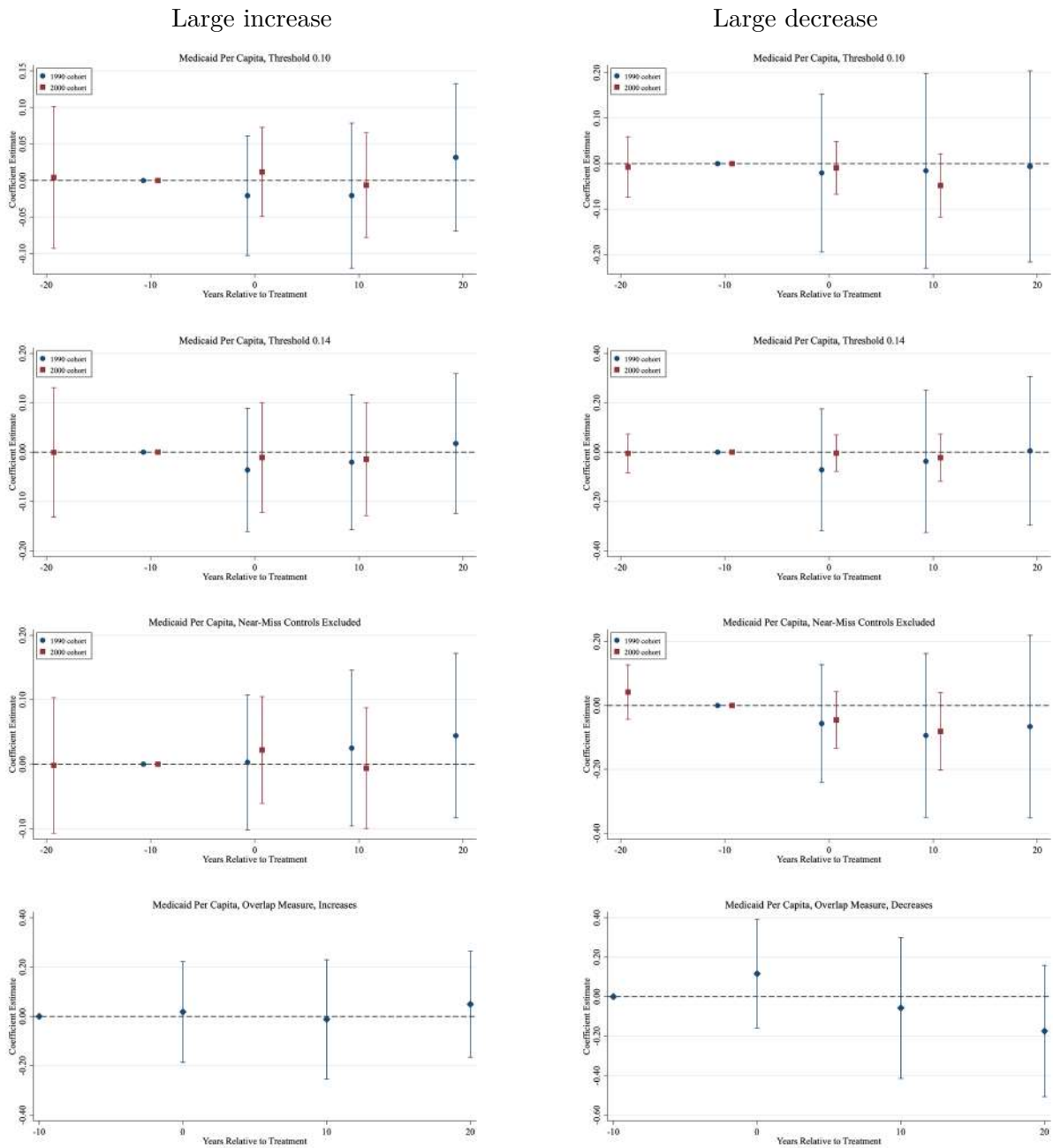
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A9: Robustness to alternative treatment definitions: log Medicare per capita



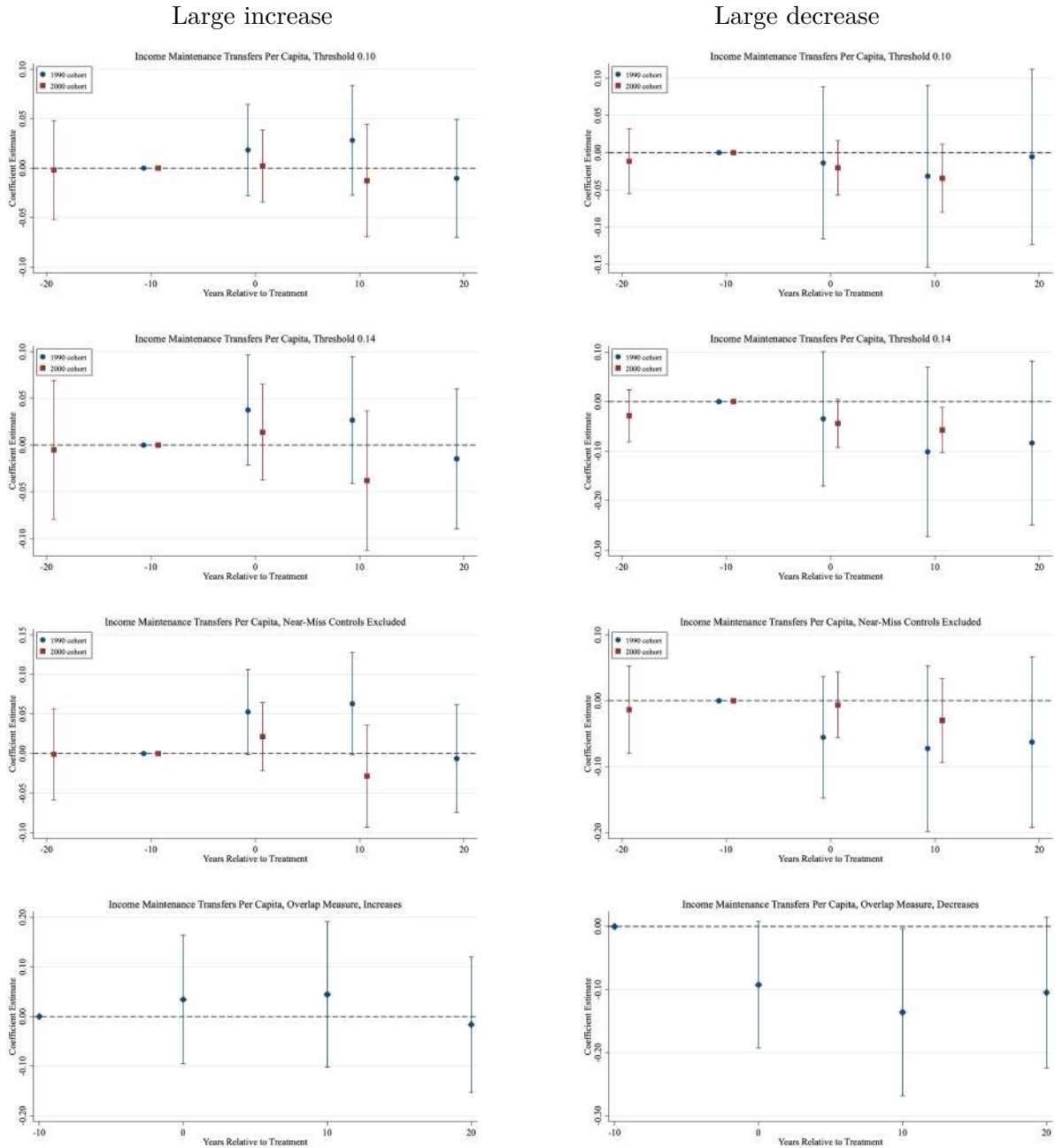
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A10: Robustness to alternative treatment definitions: log Medicaid per capita



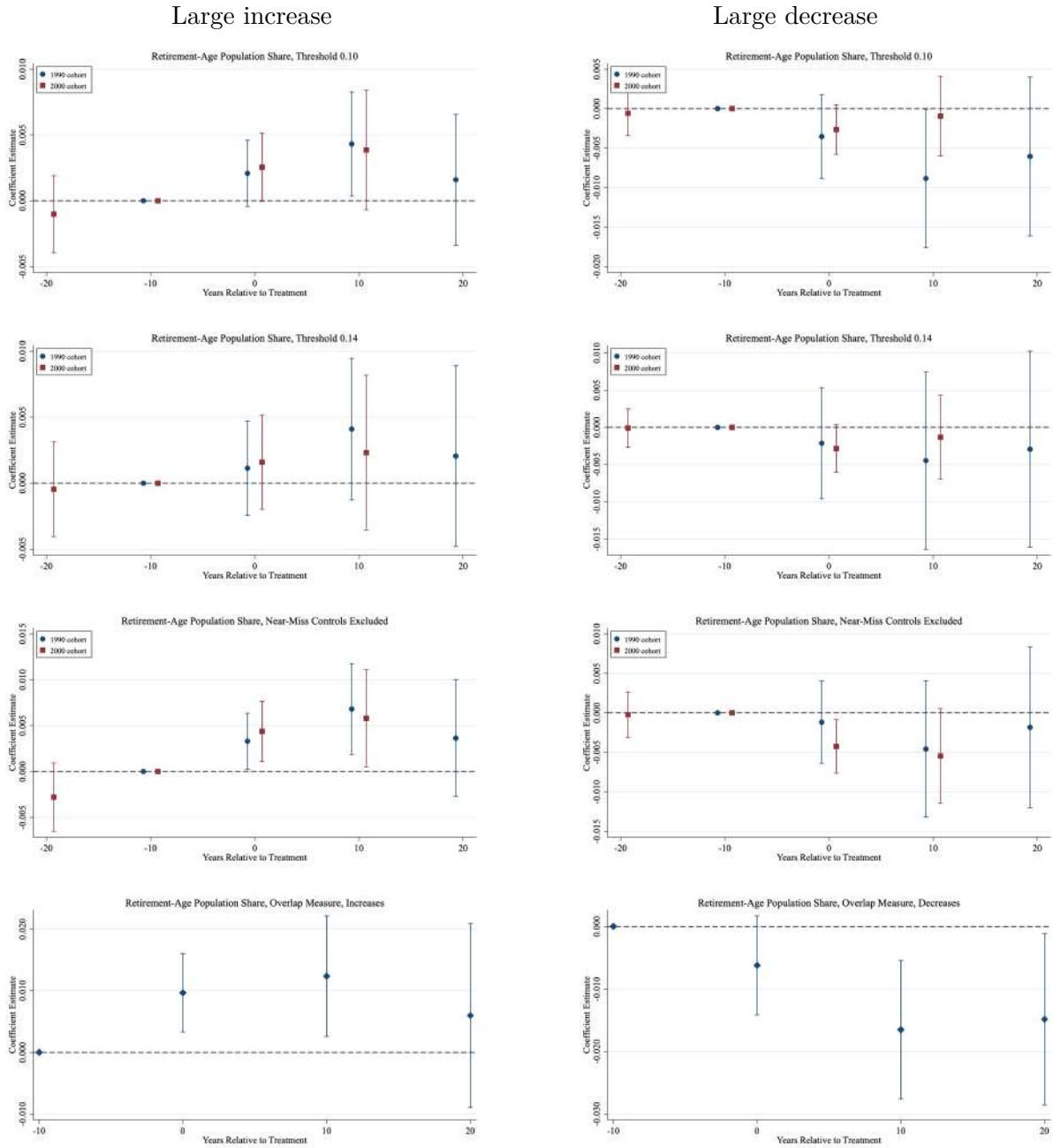
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A11: Robustness to alternative treatment definitions: log income maintenance transfers per capita



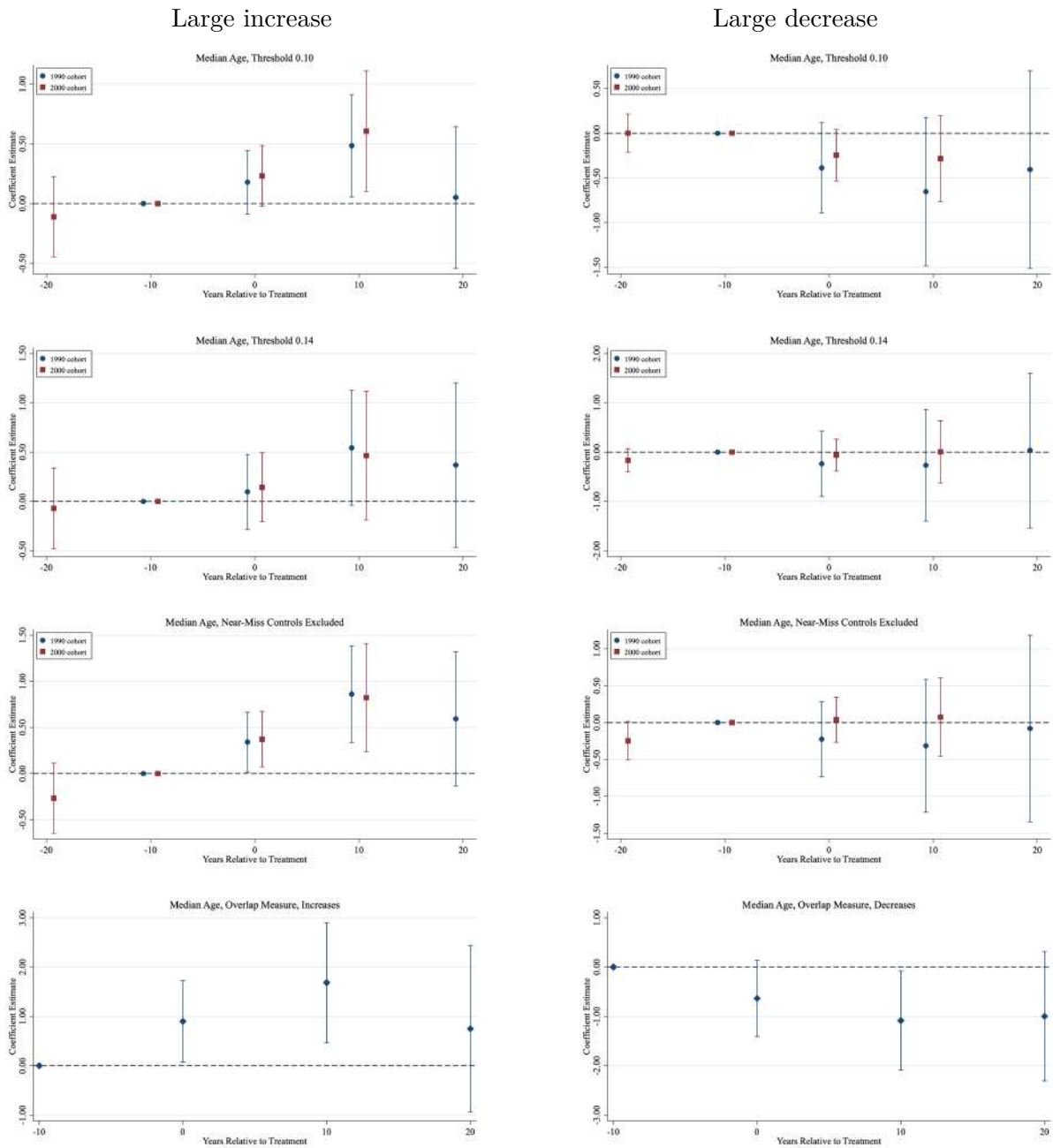
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A12: Robustness to alternative treatment definitions: the retirement-age population share



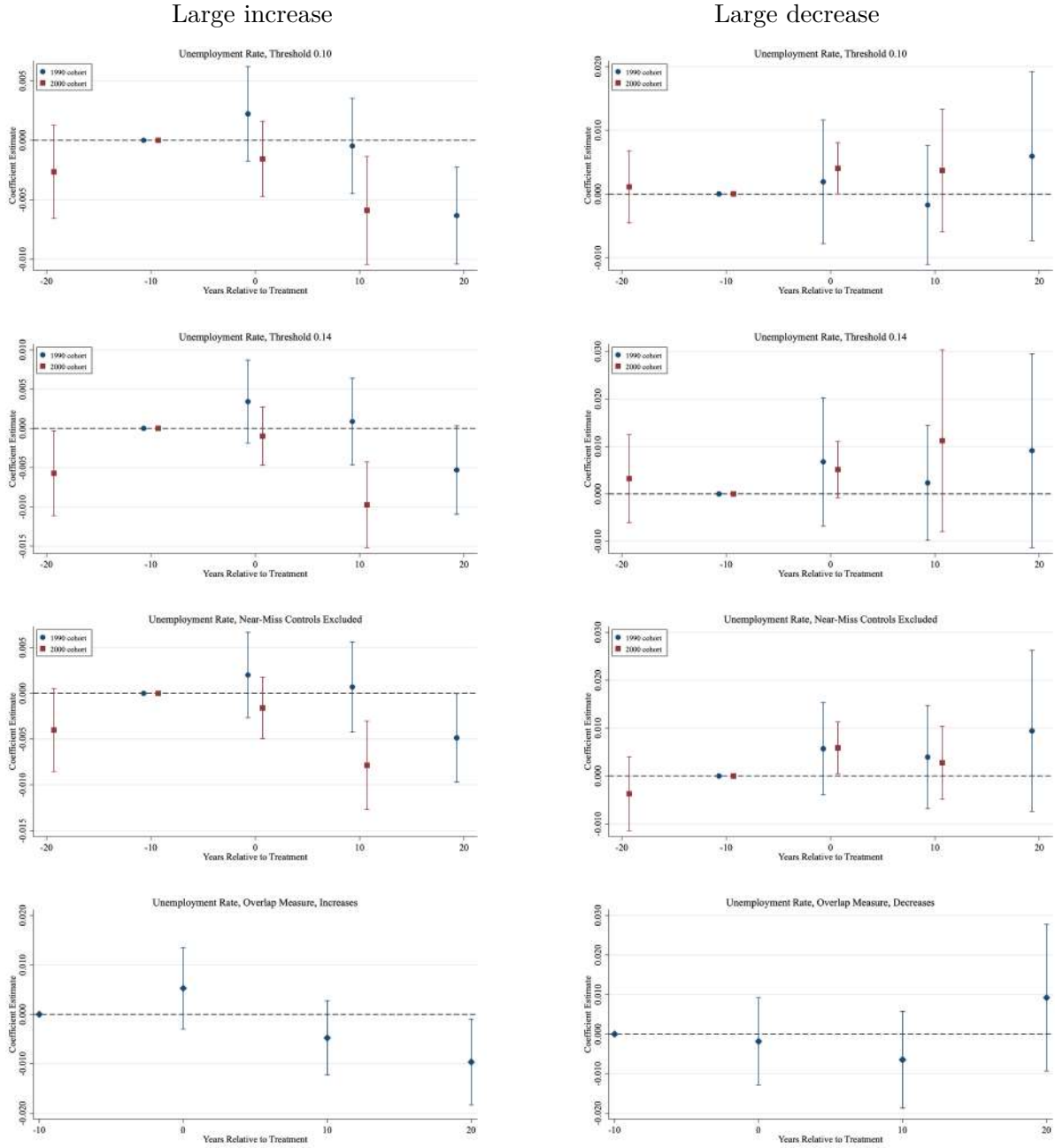
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A13: Robustness to alternative treatment definitions: median age



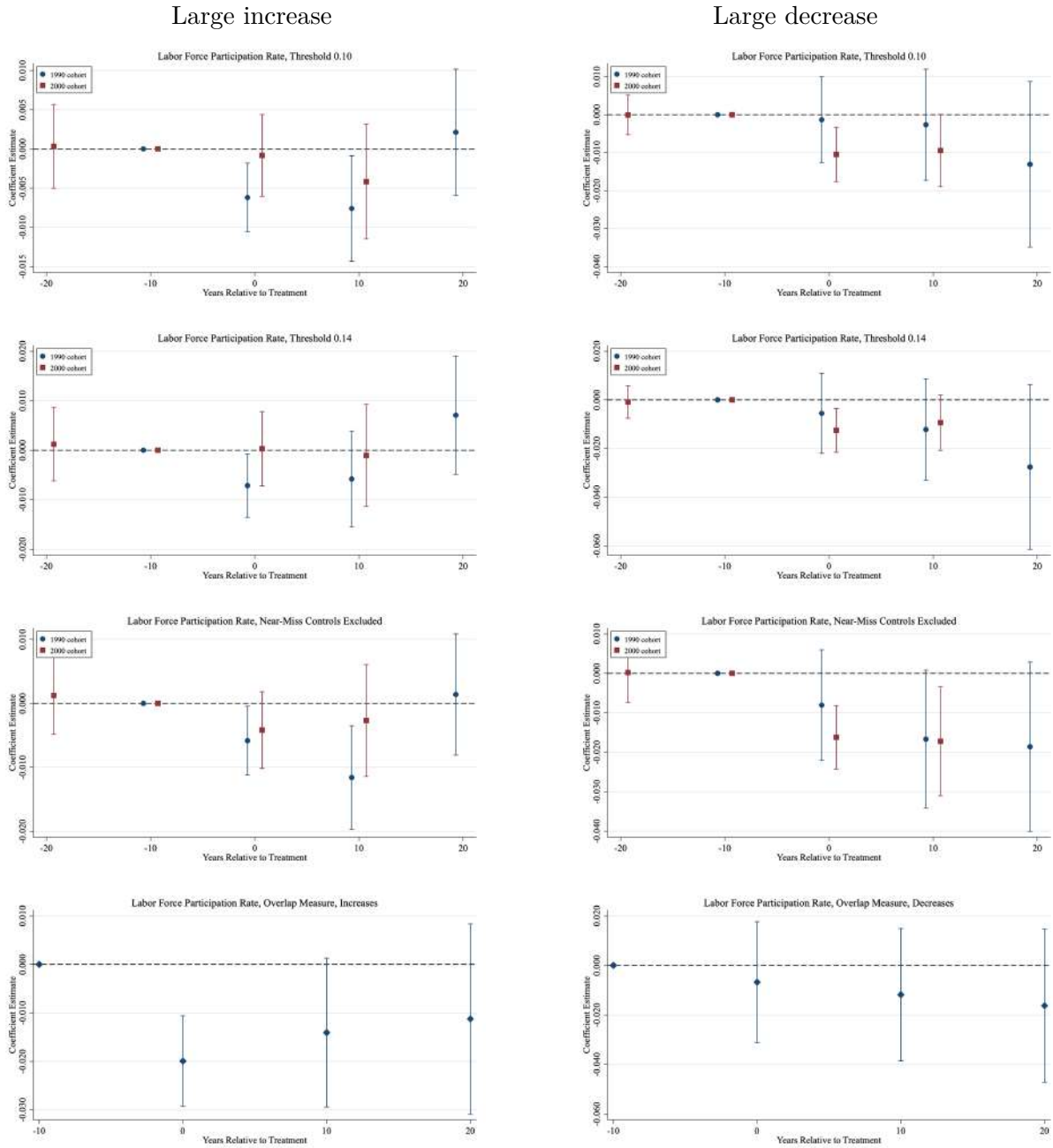
Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

Figure A14: Robustness to alternative treatment definitions: the unemployment rate



Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

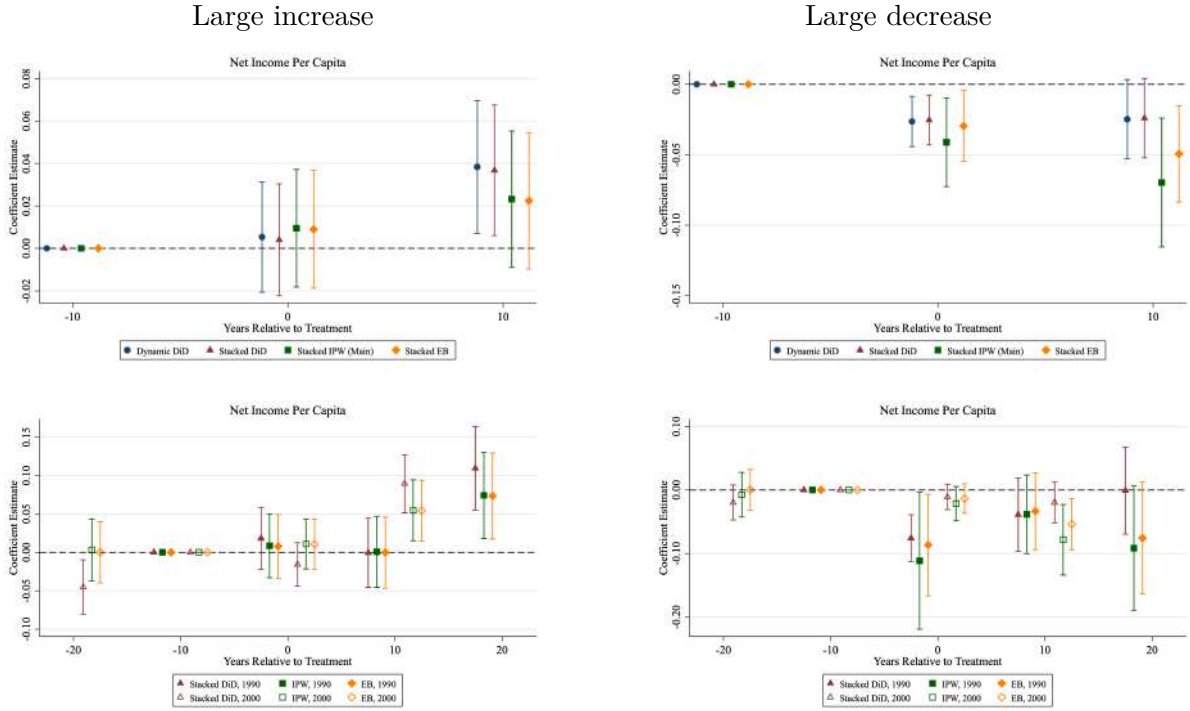
Figure A15: Robustness to alternative treatment definitions: the labor force participation rate



Notes: 95% confidence intervals. Rows: (1) threshold 0.10; (2) threshold 0.14; (3) main threshold, excluding controls whose share changed by more than 0.12 without persisting; (4) overlap measure (Section 2), 1990 cohort only. Rows (1)–(3) are cohort-specific estimates. Specifications otherwise match the main text. Standard errors clustered by county.

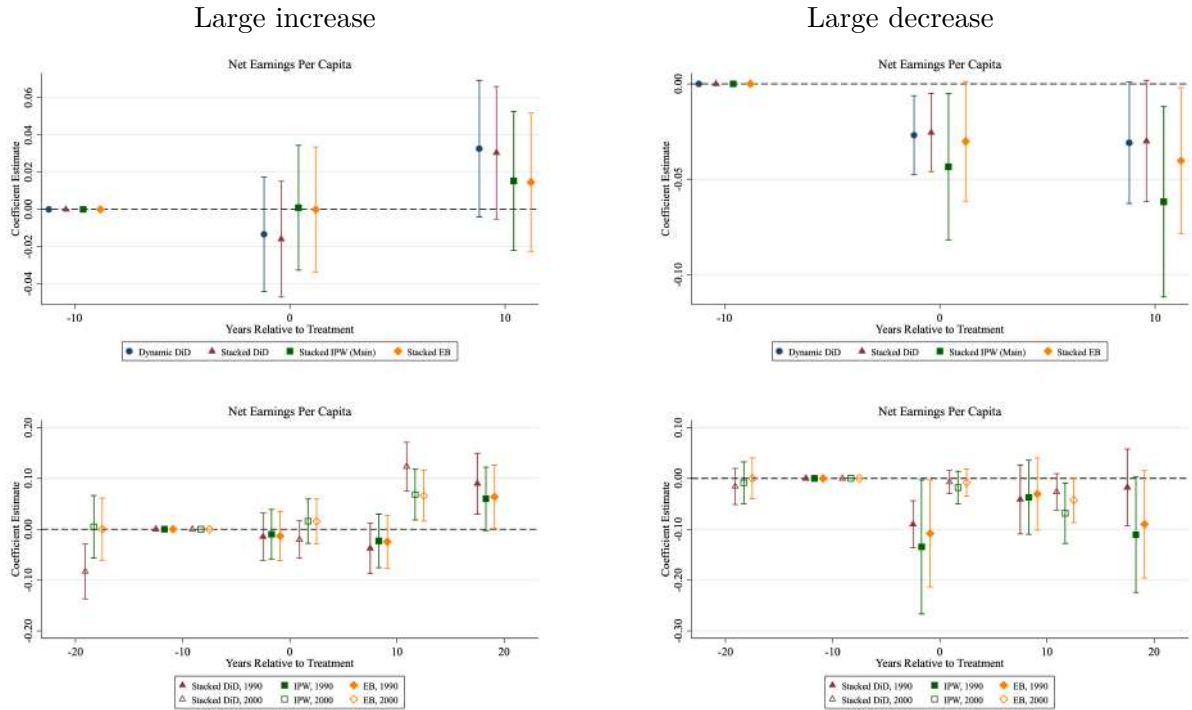
B Alternative Difference-in-Difference Estimates

Figure B1: Alternative difference-in-differences estimators: log net income per capita



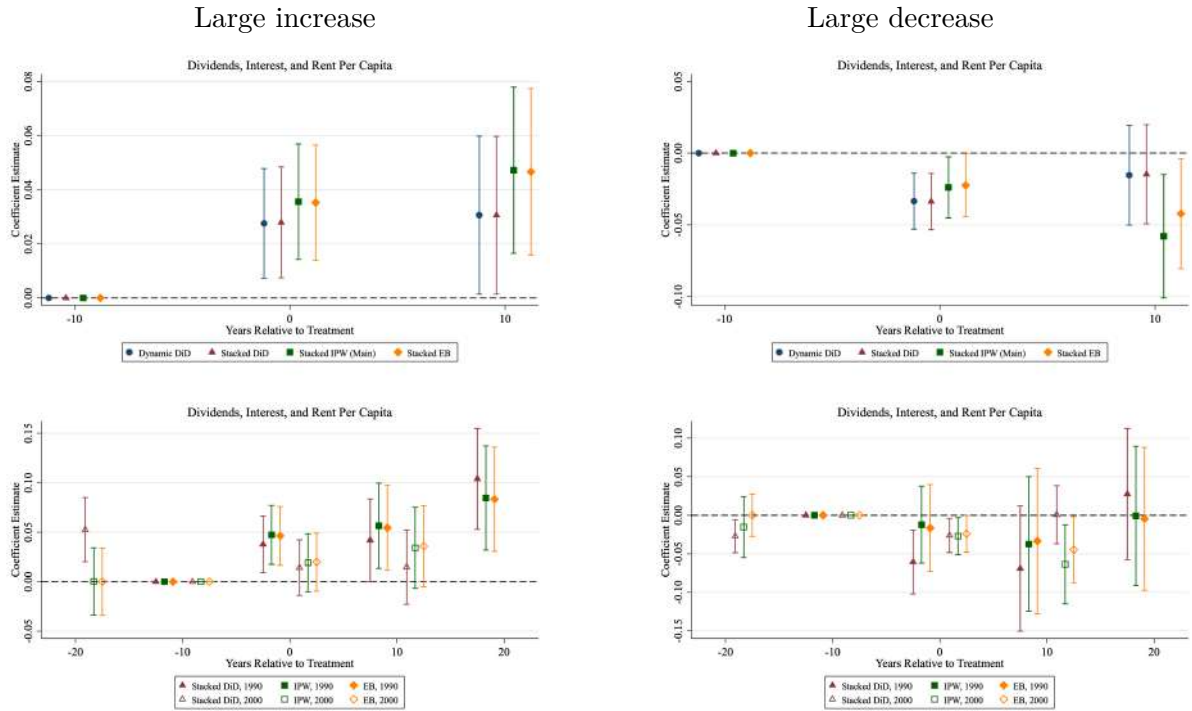
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B2: Alternative difference-in-differences estimators: log net earnings per capita



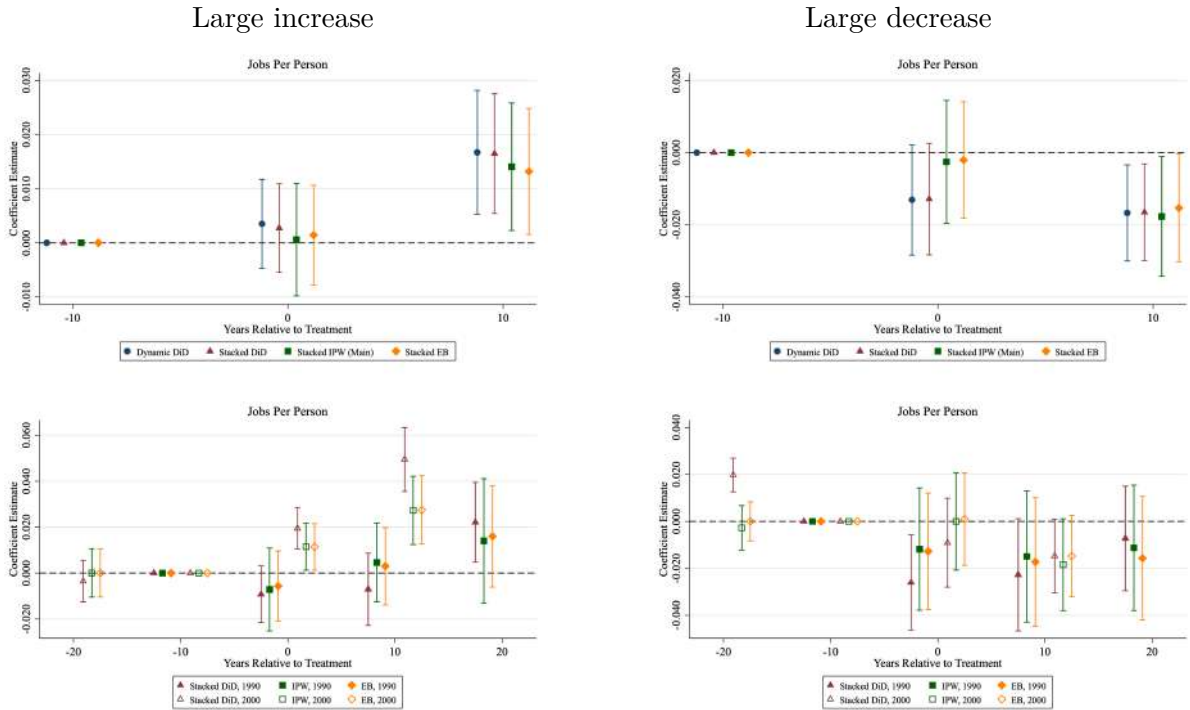
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B3: Alternative difference-in-differences estimators: log dividends, interest, and rent per capita



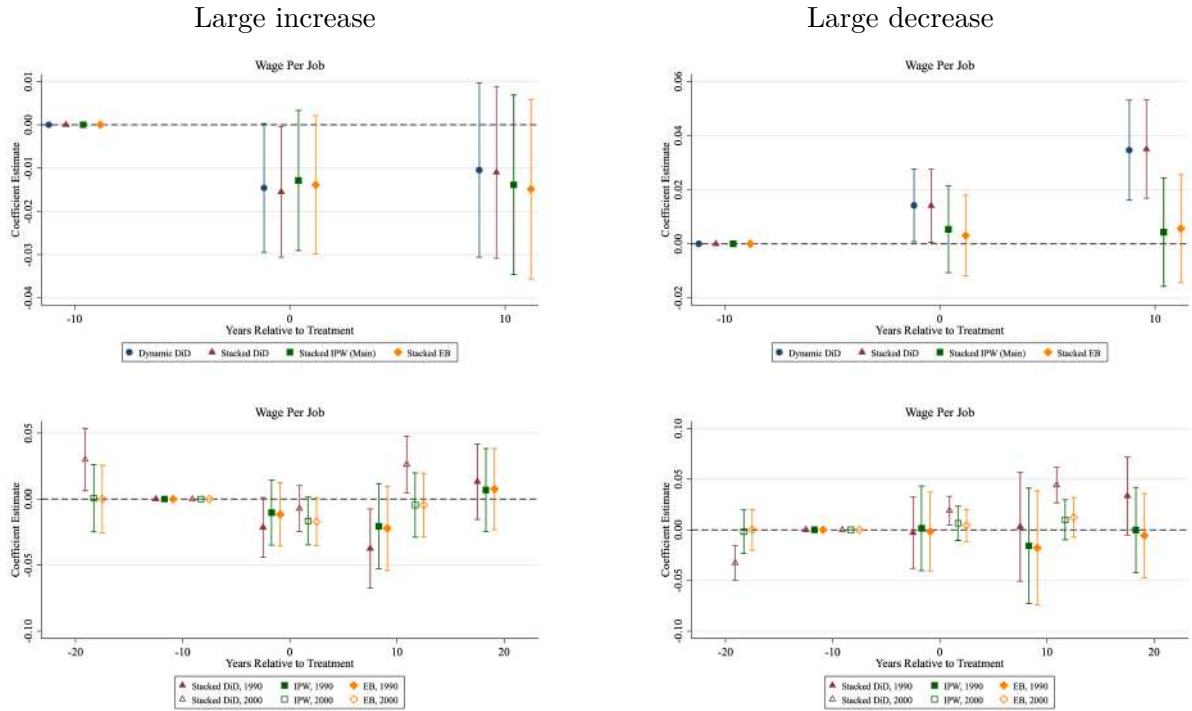
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B4: Alternative difference-in-differences estimators: jobs per person



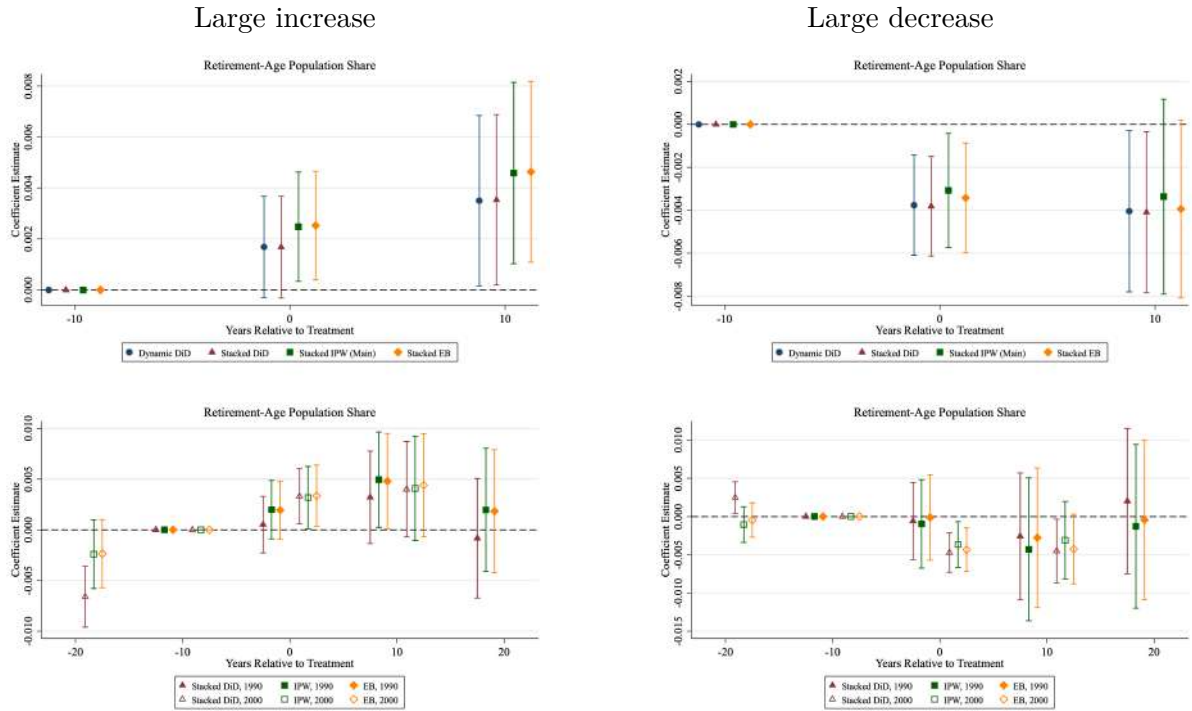
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B5: Alternative difference-in-differences estimators: log wage per wage and salary job



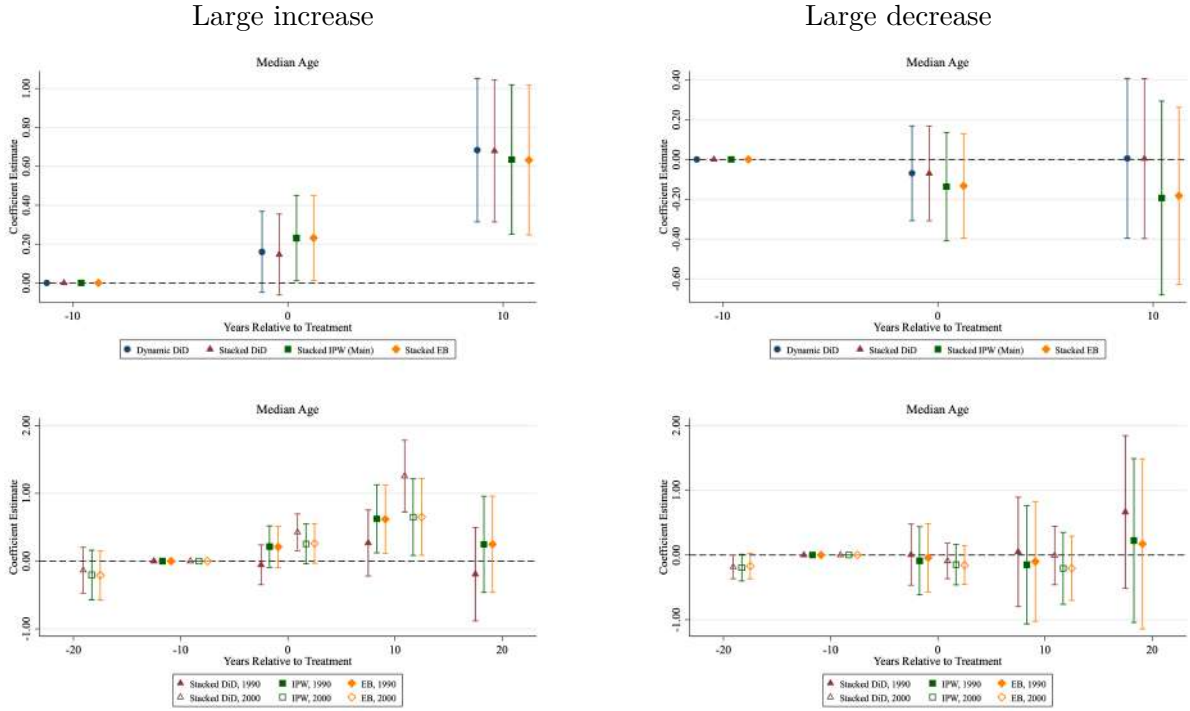
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B6: Alternative difference-in-differences estimators: the retirement-age population share



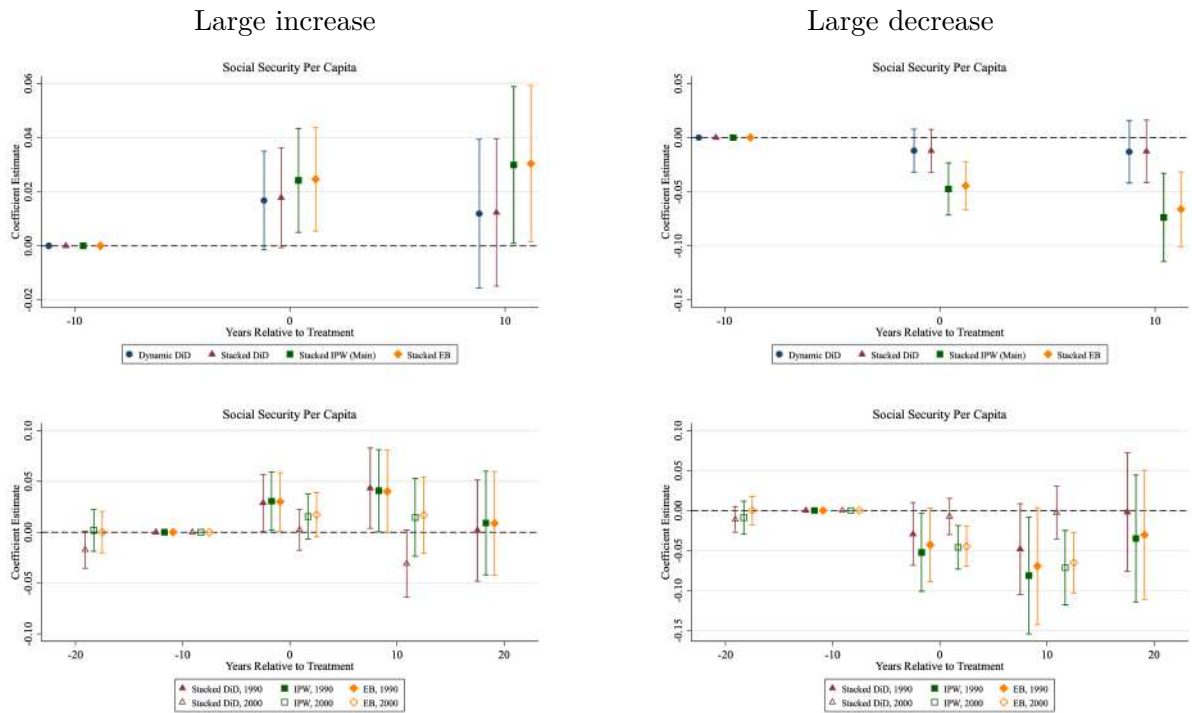
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B7: Alternative difference-in-differences estimators: median age



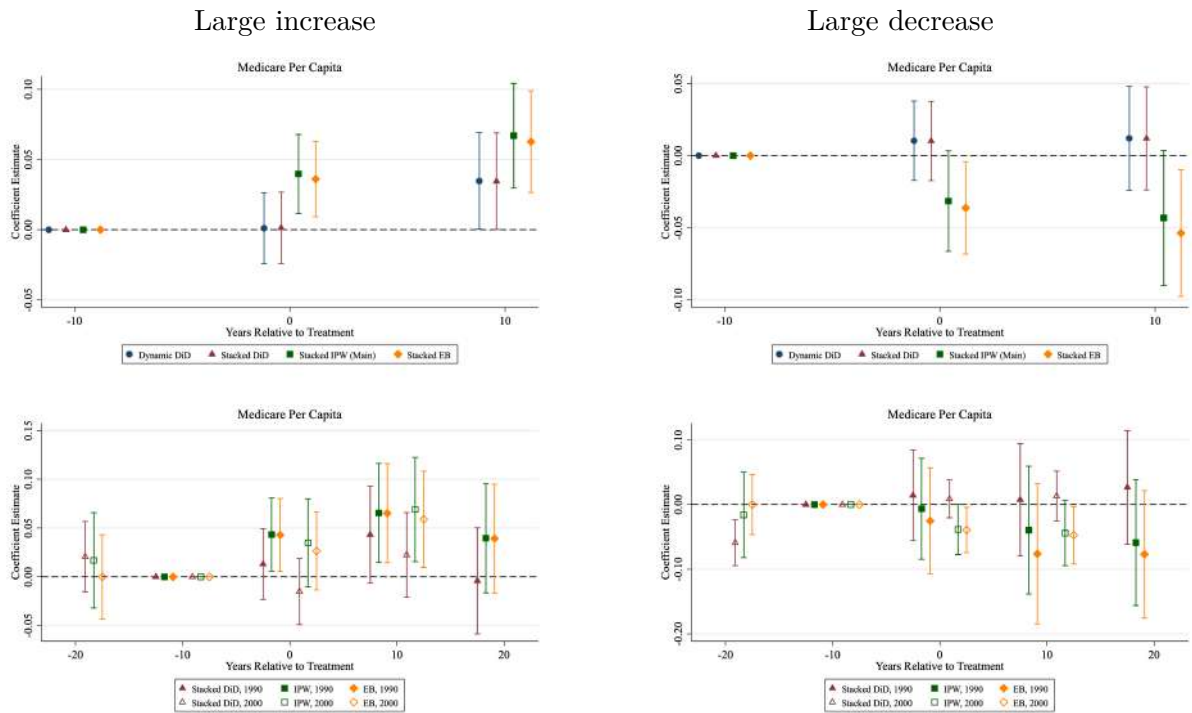
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B8: Alternative difference-in-differences estimators: log Social Security per capita



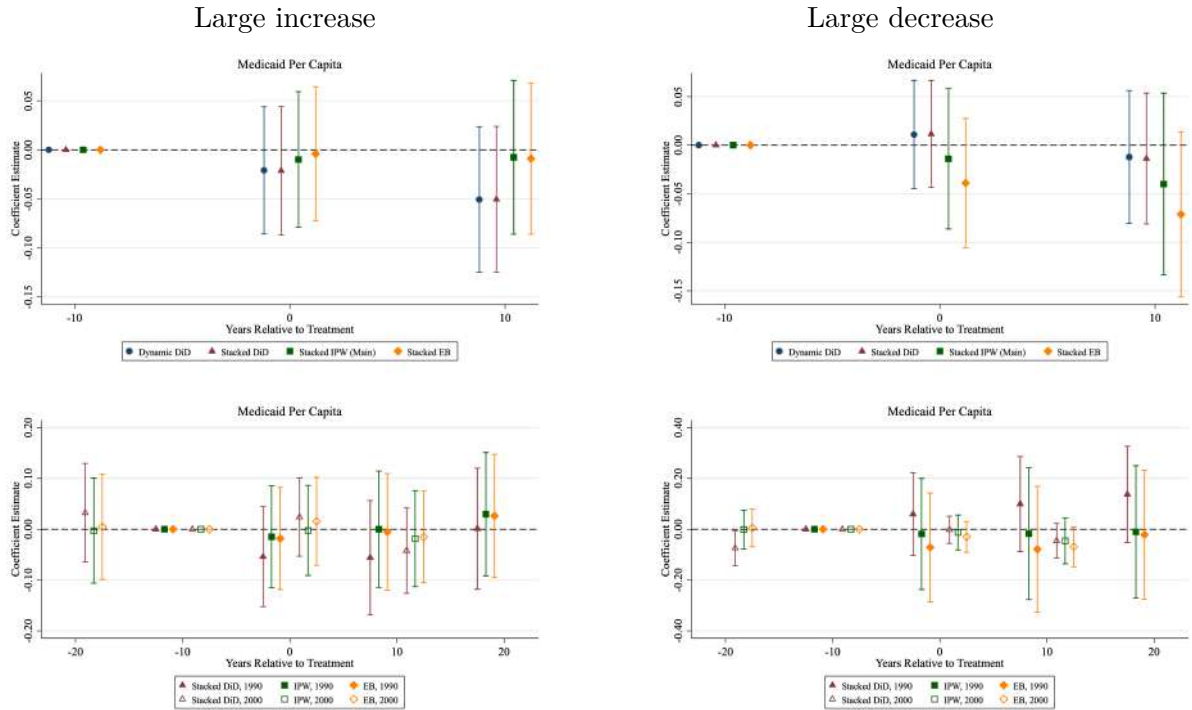
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B9: Alternative difference-in-differences estimators: log Medicare per capita



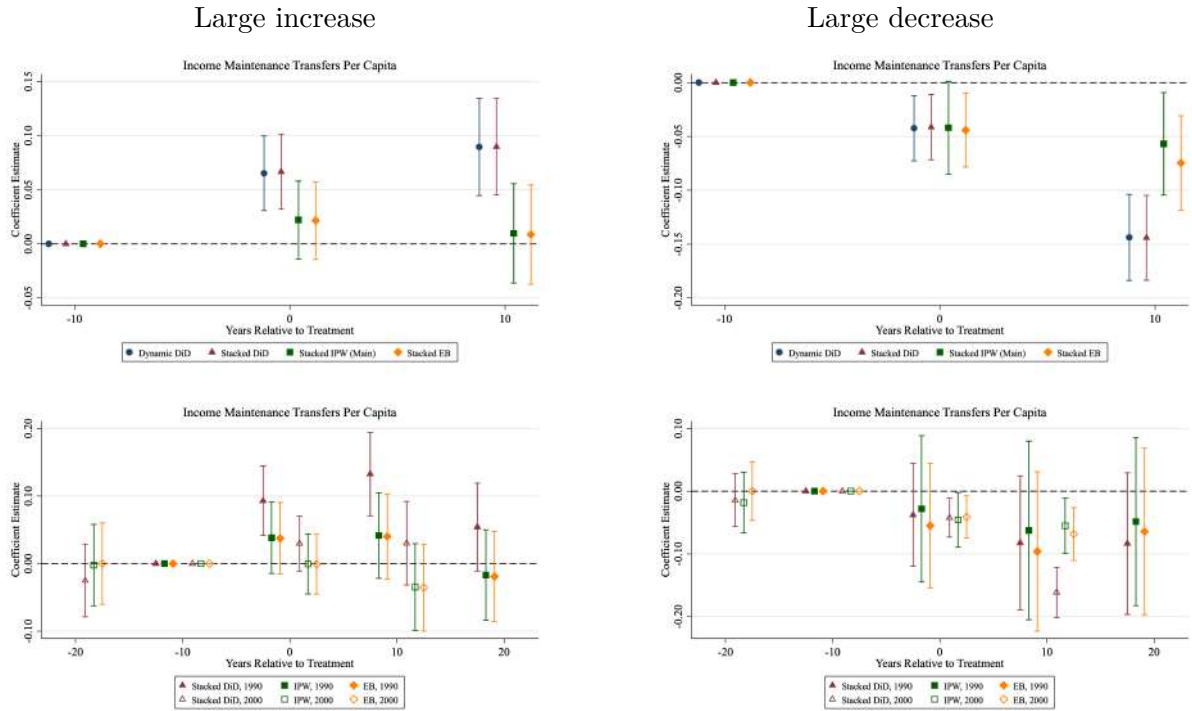
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B10: Alternative difference-in-differences estimators: log Medicaid per capita



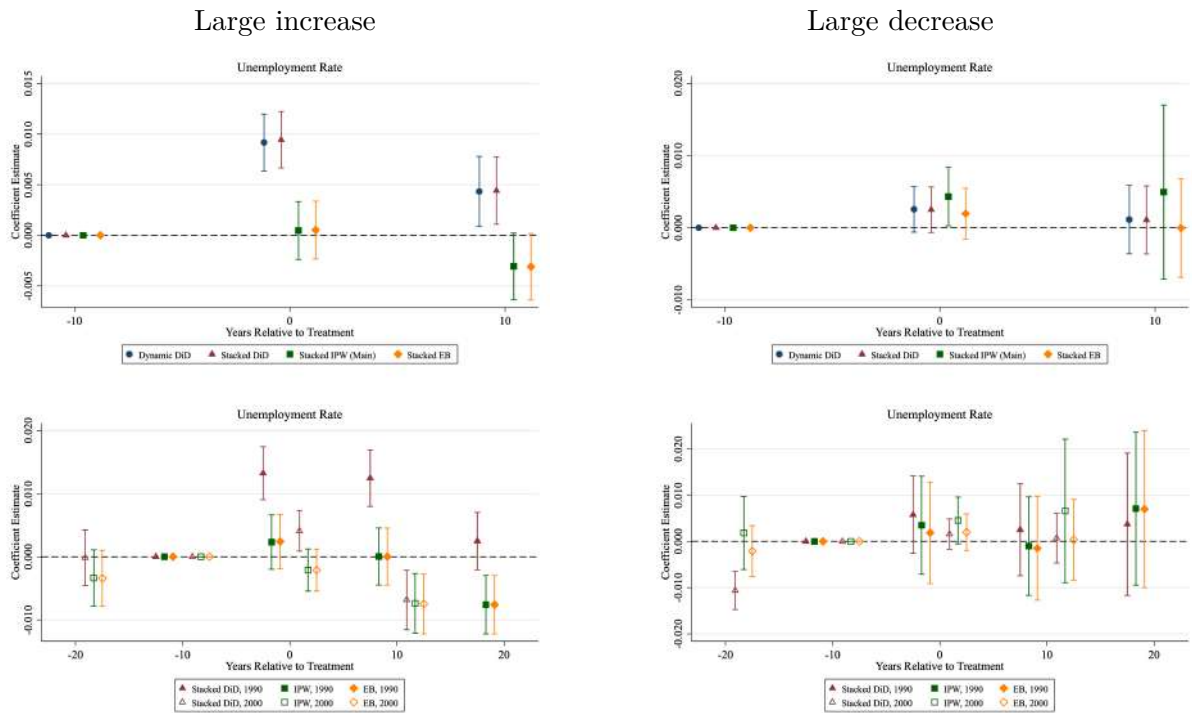
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B11: Alternative difference-in-differences estimators: log income maintenance transfers per capita



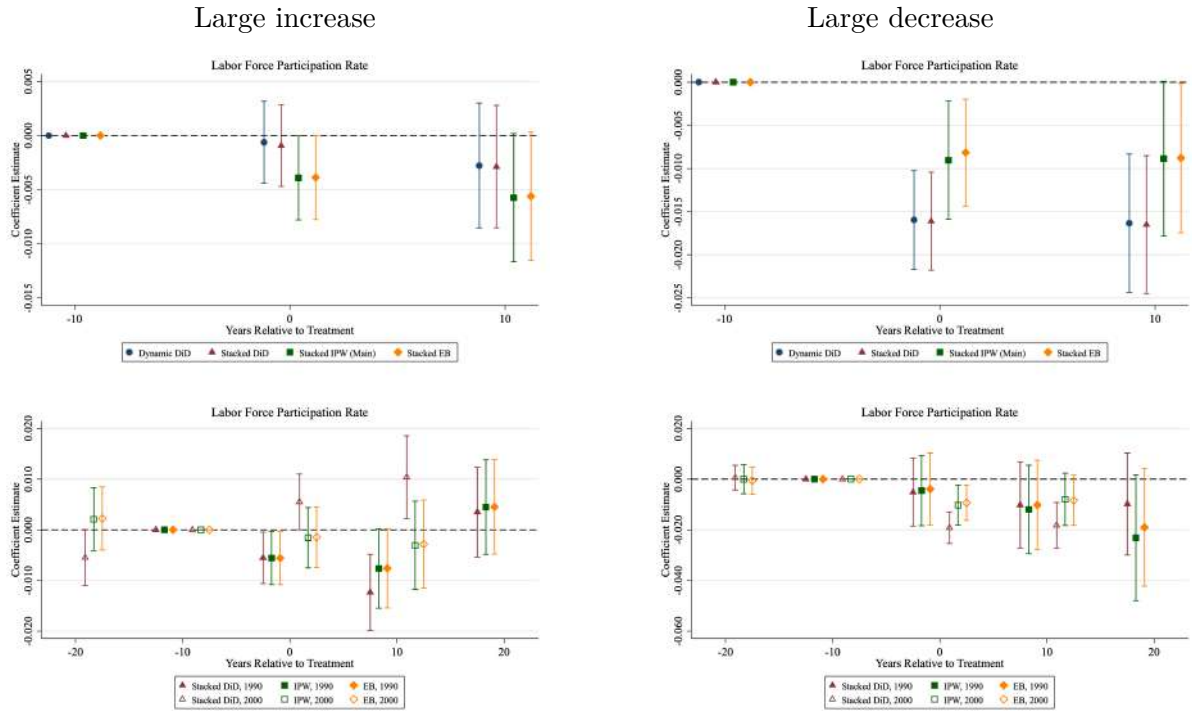
Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B12: Alternative difference-in-differences estimators: the unemployment rate



Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

Figure B13: Alternative difference-in-differences estimators: the labor force participation rate



Notes: 95% confidence intervals. Top row: pooled estimates on the common window from four estimators: two-way fixed effects on the unstacked panel; stacked difference-in-differences with corrective weights only (Wing et al., 2024); with normalized inverse propensity score weights (main specification); and with entropy balancing weights. Bottom row: each cohort on its full event window, estimated with corrective weights only, inverse propensity score weights, and entropy balancing weights; solid markers are the 1990 cohort and hollow markers the 2000 cohort. Reweighted specifications use counties with complete matching covariates. Standard errors clustered by county.

C Additional Summary Statistics and Balance Tables

Table C1: Baseline Summary Statistics, Negative Treatments

	Mean	SD
<i>Main outcome: Productive income per-capita</i>		
Net income per capita (logged)	10.066	0.283
<i>Key matching covariates (lagged by 10)</i>		
Net income per capita (logged)	9.863	0.280
Adherent share	0.576	0.189
Young population %	0.231	0.030
Teen population %	0.100	0.022
Working age population %	0.528	0.035
Male population %	0.491	0.016
Urban indicator	0.318	0.466
High school % of adults	0.649	0.121
College degree % of adults	0.123	0.058
Military % of employment	0.008	0.038
Unemployment rate	0.069	0.032
Labor force participation rate	0.595	0.067
<i>Secondary outcomes</i>		
Net earnings per capita (logged)	9.784	0.304
Dividends, interest, and rent per capita (logged)	8.621	0.348
Jobs per person	0.500	0.143
Wage per job (logged)	10.354	0.210
Retirement-age population share	0.148	0.042
Median age (years)	35.860	3.989
Government transfers per capita (logged)	8.410	0.289
Social Security per capita (logged)	7.589	0.290
Medicare per capita (logged)	6.760	0.388
Medicaid per capita (logged)	6.432	0.661
Income maintenance per capita (logged)	5.988	0.534

N=5,156 county-stack observations: all negative treatments and never-treated controls (5,154 for the retirement-age share and median age). Secondary outcomes are measured in the baseline year. Counties with a positive treatment are excluded. All covariates are lagged 10 years prior to the event for each stack; the matching covariates also include net income per capita lagged 20 years, which is not shown.

Table C2: Unweighted Difference-in-Means in $T-10$: Treated (Negative Jumps) vs Control (Never Treated)

	Control	Treated	Difference	P-value
<i>Matching covariates</i>				
Net income per capita (logged)	9.867	9.774	-0.093	0.000
Adherent share	0.567	0.777	0.210	0.000
Young population %	0.230	0.240	0.009	0.000
Teen population %	0.100	0.096	-0.004	0.005
Working age population %	0.529	0.523	-0.005	0.028
Male population %	0.491	0.490	-0.001	0.380
Urban indicator	0.325	0.188	-0.136	0.000
High school % of adults	0.652	0.601	-0.051	0.000
College degree % of adults	0.124	0.104	-0.021	0.000
Military % of employment	0.008	0.005	-0.003	0.189
Unemployment rate	0.068	0.075	0.007	0.001
Labor force participation rate	0.595	0.583	-0.012	0.005
<i>Secondary outcomes (lagged by 10)</i>				
Net earnings per capita (logged)	9.580	9.489	-0.090	0.000
Dividends, interest, and rent per capita (logged)	8.427	8.335	-0.093	0.000
Jobs per person	0.462	0.410	-0.052	0.000
Wage per job (logged)	10.260	10.171	-0.090	0.000
Retirement-age population share	0.141	0.141	-0.000	0.936
Median age (years)	32.744	32.758	0.015	0.957
Government transfers per capita (logged)	8.101	8.174	0.073	0.000
Social Security per capita (logged)	7.351	7.319	-0.031	0.134
Medicare per capita (logged)	6.215	6.357	0.142	0.000
Medicaid per capita (logged)	5.654	5.955	0.302	0.000
Income maintenance per capita (logged)	5.753	6.127	0.373	0.000

Treated $N = 240$, Control $N = 4,916$. Difference is treated minus control. Secondary outcomes enter the matching set only in their own specifications, through their own lags. P -values from two-sided t-tests.

Table C3: Standardized Differences in Matching Covariates Before and After Weighting

	Positive treatments			Negative treatments		
	Unweighted	IPW	EB	Unweighted	IPW	EB
Net income per capita (logged)	0.042	0.003	0.000	-0.331	0.062	-0.013
Net income per capita (logged), $T - 20$	-0.006	0.009	0.000	-0.395	0.011	-0.012
Adherent share	-0.004	0.004	0.000	0.994	-0.160	-0.003
Young population %	0.053	-0.004	0.000	0.281	-0.078	0.000
Teen population %	-0.094	-0.003	0.000	-0.225	-0.050	0.000
Working age population %	-0.251	0.009	0.000	-0.150	0.056	0.000
Male population %	0.105	-0.007	0.000	-0.053	-0.085	0.000
Urban indicator	-0.077	0.001	0.000	-0.348	0.071	0.000
High school % of adults	0.037	0.006	0.000	-0.471	-0.014	0.000
College degree % of adults	0.094	0.004	0.000	-0.472	-0.039	0.000
Military % of employment	0.018	-0.003	0.000	-0.205	0.037	-0.001
Unemployment rate	-0.486	-0.007	0.000	0.196	-0.142	0.000
Labor force participation rate	0.002	0.012	0.000	-0.198	0.056	0.000

Treated minus control mean, divided by the unweighted treated standard deviation, in the baseline county-stack cells. IPW uses the normalized inverse propensity score weights of the main specification; EB uses entropy balancing weights. Covariates are lagged 10 years unless noted.

D Additional Results

Table D1: Pooled Stacked Difference-in-Differences Estimates, All Outcomes

	Large increase		Large decrease	
	t	$t + 10$	t	$t + 10$
<i>Net income and components</i>				
Net income per capita (log)	0.009 (0.014)	0.023 (0.016)	-0.041** (0.016)	-0.070*** (0.023)
Net earnings per capita (log)	0.001 (0.017)	0.015 (0.019)	-0.043** (0.020)	-0.062** (0.025)
Dividends, interest, and rent per capita (log)	0.036*** (0.011)	0.047*** (0.016)	-0.024** (0.011)	-0.058*** (0.022)
Wage per wage and salary job (log)	-0.013 (0.008)	-0.014 (0.011)	0.005 (0.008)	0.004 (0.010)
Jobs per person	0.001 (0.005)	0.014** (0.006)	-0.003 (0.009)	-0.018** (0.008)
<i>Transfers per capita (log)</i>				
Government transfers	0.029*** (0.008)	0.038*** (0.012)	-0.035*** (0.010)	-0.045*** (0.016)
Social Security	0.024** (0.010)	0.030** (0.015)	-0.047*** (0.012)	-0.074*** (0.021)
Medicare	0.040*** (0.014)	0.067*** (0.019)	-0.032* (0.018)	-0.043* (0.024)
Medicaid	-0.010 (0.035)	-0.008 (0.040)	-0.014 (0.037)	-0.040 (0.048)
Income maintenance	0.022 (0.018)	0.010 (0.023)	-0.042* (0.022)	-0.057** (0.024)
Unemployment insurance	0.004 (0.044)	0.006 (0.047)	-0.057 (0.045)	0.003 (0.064)
Veterans' benefits	-0.001 (0.017)	-0.004 (0.027)	-0.009 (0.029)	-0.027 (0.034)
<i>Demographic composition</i>				
Retirement-age population share	0.002** (0.001)	0.005** (0.002)	-0.003** (0.001)	-0.003 (0.002)
Median age (years)	0.231** (0.112)	0.634*** (0.196)	-0.137 (0.139)	-0.194 (0.249)
<i>Labor market and education</i>				
Unemployment rate	0.000 (0.001)	-0.003* (0.002)	0.004** (0.002)	0.005 (0.006)
Labor force participation rate	-0.004** (0.002)	-0.006* (0.003)	-0.009*** (0.003)	-0.009* (0.005)
High school share of adults	-0.004 (0.002)	-0.004 (0.003)	-0.005 (0.004)	-0.006 (0.006)
College-degree share of adults	0.002 (0.002)	0.002 (0.003)	0.001 (0.002)	0.000 (0.003)

Notes: Each row reports the pooled stacked difference-in-differences estimates at treatment (t) and ten years later ($t + 10$), relative to $t - 10$, on the event window common to both cohorts, with county-by-stack and year-by-stack fixed effects and normalized inverse propensity score weights. Log outcomes are in log points; the retirement-age share, jobs per person, unemployment, participation, and education are in levels (shares or rates). Matching covariates are those in Table 3 plus income at $T - 20$; outcomes from BEA data also include two lags of the outcome, and the retirement-age share and median age include the ten-year lag of the outcome. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D2: Estimates for Figure 2:
net income per capita

	Increase	Decrease
<i>Pooled, common window</i>		
$k = 0$	0.009 (0.014)	-0.041** (0.016)
$k = 10$	0.023 (0.016)	-0.070*** (0.023)
Observations	15,498	15,459
<i>1990 cohort</i>		
$k = 0$	0.008 (0.021)	-0.111** (0.055)
$k = 10$	0.001 (0.024)	-0.038 (0.031)
$k = 20$	0.074*** (0.029)	-0.092* (0.050)
Observations	10,416	10,036
<i>2000 cohort</i>		
$k = -20$	0.003 (0.021)	-0.008 (0.018)
$k = 0$	0.011 (0.017)	-0.022 (0.014)
$k = 10$	0.055*** (0.020)	-0.078*** (0.028)
Observations	10,248	10,576

Notes: Log net income per capita. Matching covariates as in Table 3 plus income at $T - 20$. Event time k is years relative to treatment; $k = -10$ is the reference period. Estimates use normalized inverse propensity score weights; cohort-specific estimates include county and year fixed effects within the stack. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D3: Estimates for Figure 3: components of net income

	Net earnings		Dividends, interest, rent	
	Increase	Decrease	Increase	Decrease
<i>1990 cohort</i>				
$k = 0$	-0.010 (0.025)	-0.135** (0.067)	0.047*** (0.015)	-0.012 (0.026)
$k = 10$	-0.023 (0.027)	-0.037 (0.037)	0.057** (0.022)	-0.037 (0.045)
$k = 20$	0.060* (0.032)	-0.111* (0.058)	0.085*** (0.027)	-0.001 (0.046)
Observations	10,400	10,028	10,416	10,036
<i>2000 cohort</i>				
$k = -20$	0.005 (0.031)	-0.009 (0.021)	0.000 (0.017)	-0.015 (0.020)
$k = 0$	0.016 (0.022)	-0.018 (0.016)	0.019 (0.015)	-0.027** (0.012)
$k = 10$	0.068*** (0.025)	-0.069** (0.030)	0.034 (0.021)	-0.064** (0.026)
Observations	10,240	10,568	10,248	10,576

Notes: Logs per capita. Matching covariates also include two lags of the outcome. Event time k is years relative to treatment; $k = -10$ is the reference period. Estimates use normalized inverse propensity score weights; cohort-specific estimates include county and year fixed effects within the stack. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D4: Estimates for Figure 4: employment and pay

	Jobs per person		Wage per job (log)	
	Increase	Decrease	Increase	Decrease
<i>1990 cohort</i>				
$k = 0$	-0.007 (0.009)	-0.012 (0.013)	-0.010 (0.013)	0.001 (0.021)
$k = 10$	0.005 (0.009)	-0.015 (0.014)	-0.021 (0.016)	-0.016 (0.029)
$k = 20$	0.014 (0.014)	-0.011 (0.014)	0.007 (0.016)	-0.000 (0.021)
Observations	10,416	10,036	10,416	10,036
<i>2000 cohort</i>				
$k = -20$	0.000 (0.005)	-0.003 (0.005)	0.001 (0.013)	-0.002 (0.011)
$k = 0$	0.011** (0.005)	0.000 (0.011)	-0.017* (0.009)	0.006 (0.009)
$k = 10$	0.027*** (0.008)	-0.018* (0.010)	-0.005 (0.012)	0.010 (0.010)
Observations	10,248	10,576	10,248	10,576

Notes: Jobs per person in levels; wage per wage and salary job in logs. Matching covariates also include two lags of the outcome. Event time k is years relative to treatment; $k = -10$ is the reference period. Estimates use normalized inverse propensity score weights; cohort-specific estimates include county and year fixed effects within the stack. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D5: Estimates for Figure 5:
total government transfers

	Government transfers	
	Increase	Decrease
<i>1990 cohort</i>		
$k = 0$	0.033*** (0.012)	-0.033 (0.023)
$k = 10$	0.048*** (0.018)	-0.059** (0.029)
$k = 20$	0.022 (0.019)	-0.025 (0.032)
Observations	10,416	10,036
<i>2000 cohort</i>		
$k = -20$	0.005 (0.015)	-0.008 (0.011)
$k = 0$	0.023** (0.011)	-0.036*** (0.012)
$k = 10$	0.023 (0.015)	-0.042** (0.019)
Observations	10,248	10,576

Notes: Logs per capita. Matching covariates also include two lags of the outcome. Event time k is years relative to treatment; $k = -10$ is the reference period. Estimates use normalized inverse propensity score weights; cohort-specific estimates include county and year fixed effects within the stack. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D6: Estimates for Figure 6: Social Security and Medicare

	Social Security		Medicare	
	Increase	Decrease	Increase	Decrease
<i>1990 cohort</i>				
$k = 0$	0.030** (0.015)	-0.052** (0.025)	0.043** (0.019)	-0.006 (0.040)
$k = 10$	0.041** (0.021)	-0.081** (0.037)	0.065** (0.026)	-0.039 (0.050)
$k = 20$	0.009 (0.026)	-0.035 (0.041)	0.040 (0.029)	-0.059 (0.050)
Observations	10,416	10,036	10,416	10,036
<i>2000 cohort</i>				
$k = -20$	0.002 (0.010)	-0.009 (0.011)	0.017 (0.025)	-0.016 (0.034)
$k = 0$	0.015 (0.011)	-0.046*** (0.014)	0.035 (0.023)	-0.038* (0.020)
$k = 10$	0.015 (0.019)	-0.072*** (0.024)	0.069** (0.027)	-0.044* (0.026)
Observations	10,248	10,576	10,248	10,576

Notes: Logs per capita. Matching covariates also include two lags of the outcome. Event time k is years relative to treatment; $k = -10$ is the reference period. Estimates use normalized inverse propensity score weights; cohort-specific estimates include county and year fixed effects within the stack. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D7: Estimates for Figure 7: Medicaid and income maintenance

	Medicaid		Income maintenance	
	Increase	Decrease	Increase	Decrease
<i>1990 cohort</i>				
$k = 0$	-0.015 (0.051)	-0.019 (0.111)	0.038 (0.027)	-0.028 (0.060)
$k = 10$	-0.000 (0.058)	-0.018 (0.133)	0.042 (0.032)	-0.063 (0.073)
$k = 20$	0.030 (0.062)	-0.010 (0.133)	-0.017 (0.034)	-0.049 (0.068)
Observations	10,300	9,924	10,416	10,036
<i>2000 cohort</i>				
$k = -20$	-0.003 (0.053)	-0.001 (0.039)	-0.002 (0.031)	-0.018 (0.025)
$k = 0$	-0.002 (0.045)	-0.013 (0.035)	-0.000 (0.023)	-0.046** (0.022)
$k = 10$	-0.019 (0.048)	-0.046 (0.046)	-0.035 (0.033)	-0.055** (0.022)
Observations	10,164	10,512	10,248	10,576

Notes: Logs per capita. Matching covariates also include two lags of the outcome. Event time k is years relative to treatment; $k = -10$ is the reference period. Estimates use normalized inverse propensity score weights; cohort-specific estimates include county and year fixed effects within the stack. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D8: Estimates for Figure 8: demographic composition

	Retirement-age share		Median age	
	Increase	Decrease	Increase	Decrease
<i>1990 cohort</i>				
$k = 0$	0.002 (0.001)	-0.001 (0.003)	0.215 (0.156)	-0.089 (0.269)
$k = 10$	0.005** (0.002)	-0.004 (0.005)	0.624** (0.256)	-0.152 (0.466)
$k = 20$	0.002 (0.003)	-0.001 (0.005)	0.247 (0.362)	0.223 (0.646)
Observations	10,416	10,036	10,416	10,036
<i>2000 cohort</i>				
$k = -20$	-0.002 (0.002)	-0.001 (0.001)	-0.204 (0.186)	-0.195* (0.105)
$k = 0$	0.003** (0.002)	-0.004** (0.002)	0.253* (0.149)	-0.150 (0.160)
$k = 10$	0.004 (0.003)	-0.003 (0.003)	0.649** (0.289)	-0.206 (0.283)
Observations	10,248	10,576	10,248	10,576

Notes: Retirement-age share in levels (share of population aged 65 and older); median age in years. Matching covariates also include the ten-year lag of the outcome. Event time k is years relative to treatment; $k = -10$ is the reference period. Estimates use normalized inverse propensity score weights; cohort-specific estimates include county and year fixed effects within the stack. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

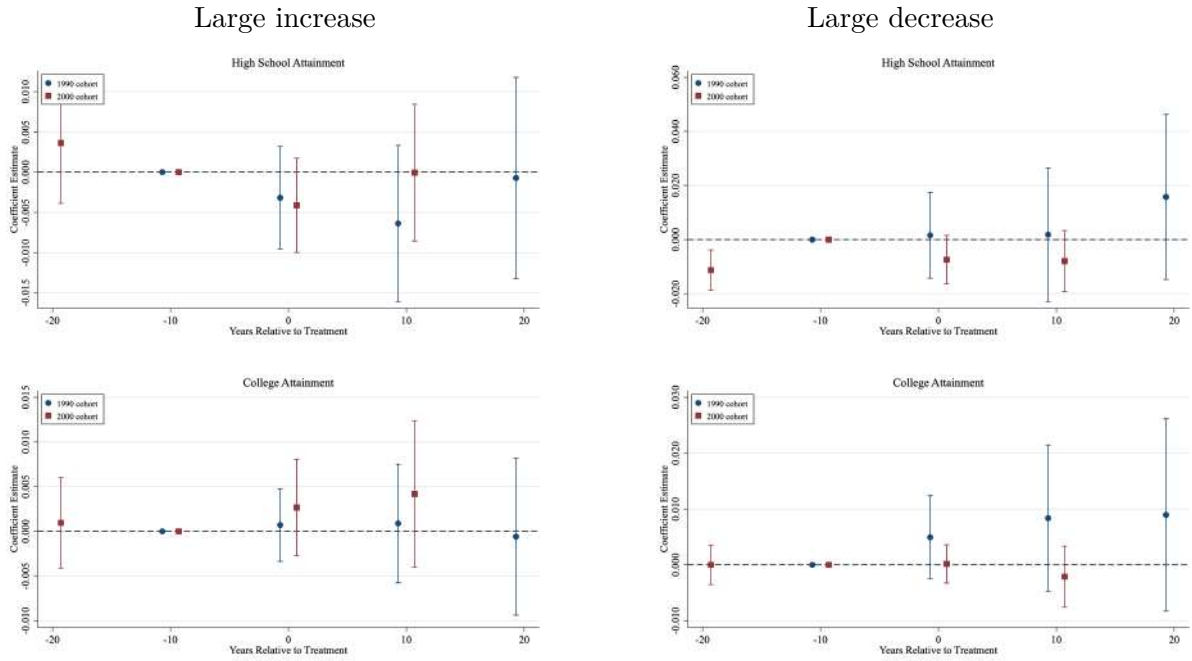
Table D9: Estimates for Figure 9: labor market outcomes

	Unemployment rate		Labor force participation	
	Increase	Decrease	Increase	Decrease
<i>1990 cohort</i>				
$k = 0$	0.002 (0.002)	0.003 (0.005)	-0.006** (0.003)	-0.005 (0.007)
$k = 10$	0.000 (0.002)	-0.001 (0.005)	-0.008* (0.004)	-0.012 (0.009)
$k = 20$	-0.008*** (0.002)	0.007 (0.008)	0.004 (0.005)	-0.023* (0.013)
Observations	10,416	10,036	10,416	10,036
<i>2000 cohort</i>				
$k = -20$	-0.003 (0.002)	0.002 (0.004)	0.002 (0.003)	-0.000 (0.003)
$k = 0$	-0.002 (0.002)	0.005* (0.003)	-0.002 (0.003)	-0.010** (0.004)
$k = 10$	-0.007*** (0.002)	0.007 (0.008)	-0.003 (0.004)	-0.008 (0.005)
Observations	10,248	10,576	10,248	10,576

Notes: Rates in levels. Matching covariates as in Table 3 plus income at $T - 20$. Event time k is years relative to treatment; $k = -10$ is the reference period. Estimates use normalized inverse propensity score weights; cohort-specific estimates include county and year fixed effects within the stack. Standard errors clustered at the county level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E Educational Attainment

Figure E1: Educational attainment



Notes: 95% confidence intervals. Top row: share of adults (25+) with only a high school diploma; bottom row: share with a college degree. Cohort-specific estimates; specifications otherwise match the main text. Standard errors clustered by county.